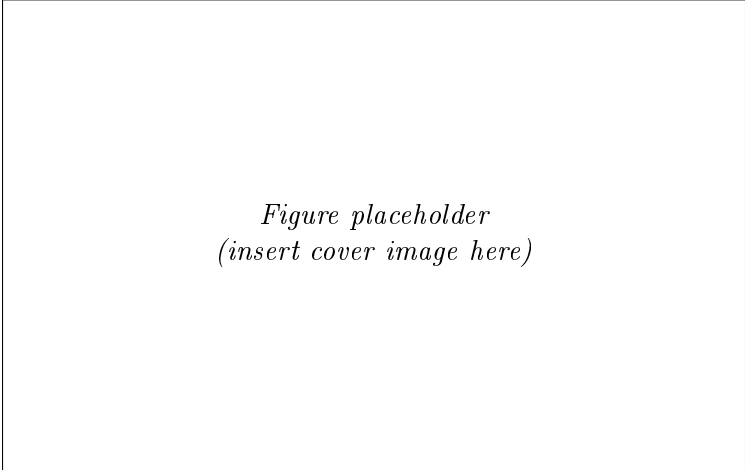


Technical Design Document

Cubli



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Space Challenges 2026

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1 Introduction

1.1 Background & Problem Statement

The Cubli is a reaction-wheel inverted pendulum: a cube-shaped platform that balances on an edge or a corner using only internal actuation, with no external supports or contact forces beyond the single line or point of contact with the ground. The concept was introduced by the Institute for Dynamic Systems and Control (IDSC) at ETH Zurich [1, 2], whose original prototype ($15 \times 15 \times 15$ cm) mounts one brushless DC motor and flywheel on each of three orthogonal faces. Commanding a change in wheel speed produces, by conservation of angular momentum, an equal and opposite reaction torque on the cube body; braking a wheel abruptly transfers its stored angular momentum to the body in a near-impulsive event, which is what allows the cube to *jump up* from resting on a face into a balance on an edge or corner.

Dynamically, the system is an *underactuated*, nonlinear, unstable-equilibrium control problem: three actuators (the wheels) must stabilize a body with more degrees of freedom than actuated inputs, about equilibria (edge and corner balance) that are open-loop unstable and increasingly coupled across axes as the contact geometry shrinks from a line (edge) to a point (corner). This is structurally the same problem as the classical inverted pendulum, extended from a single rotational axis to a fully three-dimensional, momentum-exchange-actuated body — which is why the Cubli has become a standard benchmark platform in the controls community over the past decade.

The problem this project addresses is therefore twofold: (i) derive and validate a dynamic model and controller capable of stabilizing this underactuated system about its unstable equilibria and driving it through commanded reorientations, and (ii) realize this in hardware as a modular, largely parametric design, since final actuator and sensor specifications are not yet confirmed at the time of writing. The target capability set, in order of increasing difficulty, is edge balance, corner balance, controlled rotation, and walking (Section 1.3).

Why this is a hard build, not a decorative demo. It is worth stating plainly why a machine that looks like a cube with three flywheels resists casual replication. The difficulty is not that any single subsystem is exotic — none is — but that every subsystem sits close to a physical bound, and the bounds are *coupled*: relieving one tightens another. The following failure modes recur across published and amateur replications, and each is a consequence of that coupling rather than of poor workmanship.

- **The jump-up is an energy problem with no margin.** The wheel must store, at its maximum speed, enough angular momentum to lift the cube’s center of mass over the corner. The required per-wheel inertia scales as $ML^{3/2}$ (Section 4.3.1), so mass added *anywhere* — battery, frame, fasteners — demands a larger wheel, and the wheel is itself part of M . The sizing problem is therefore a fixed-point iteration rather than a one-pass calculation, and it does not always converge: for the present cube the three-wheel set alone accounts for roughly a quarter of the total mass budget, and shrinking the wheel diameter to save space makes the situation worse, not better, because inertia must then be recovered with ballast at a smaller radius.
- **Braking torque fights the structure.** The momentum accumulated over seconds of spin-up must be shed in tens of milliseconds. That impulse passes through one motor shaft, one hub, and one motor mount — the very parts the mass budget wants thin. Many builds fail here first, and they fail in a characteristic way: the electronics are fine, the controller is sound, and the coupling, hub or mount yields on the first *successful* jump. The structural load case is set by the maneuver that proves the machine works.

- **Sensing, not the control law, sets the balancing limit.** The attitude estimate degrades from two directions at once: MEMS gyroscope bias drifts, and the accelerometer — the only absolute reference for tilt — is corrupted by vibration that the controller itself generates by spinning the wheels. Structural design and state estimation are consequently not separable concerns. A wheel half a gram out of balance at rim radius can render a perfectly correct estimator unusable, which is why wheel balance is treated here as an estimation requirement and not merely as a mechanical nicety.
- **The plant is genuinely nonlinear and constrained.** Corner balancing does not decouple into three independent single-axis problems: the axes are kinematically coupled through the contact point and the body inertia tensor. The actuators saturate in *both* torque and speed, and under any persistent disturbance the wheels drift toward their speed limit, so stored momentum must be actively managed rather than left to accumulate. Linearizing about the upright equilibrium is entirely adequate for a balancing demonstration and inadequate for anything more interesting [3].

The common thread is worth making explicit, because it determines how this project is sequenced. Replications typically fail not because the control law was wrong, but because the machine was never capable of the maneuver being asked of it: insufficient wheel inertia, motors that cannot reach their rated speed under load, a brake too slow to approximate an impulse, or a mass budget that quietly grew past the point where the stored energy suffices. Every one of these is decidable on paper, before any material is cut — and commonly is not decided there. That is the reason sizing is treated in this document as the first engineering activity rather than a detail to be settled once parts arrive.

1.2 Purpose, Application & Mission Context

The Cubli is not presented here as a balancing novelty but as a terrestrial, gravity-loaded demonstrator of **momentum-exchange attitude control** — the same physical mechanism that governs how real spacecraft point themselves. Nearly every three-axis-stabilized satellite holds and changes its attitude using reaction wheels: flywheels whose commanded speed change produces a reaction torque on the spacecraft body, exactly as in the Cubli. Spacecraft typically carry three or four such wheels (often in a pyramidal arrangement for redundancy) — directly analogous to the Cubli’s three orthogonal wheels — and this reaction-wheel approach is the standard for CubeSats and small satellites in particular, making the Cubli representative of a real, currently-flown hardware class rather than an idealized toy problem. Reaction wheels are distinct from control moment gyros (CMGs), which gimbal a constantly-spinning wheel instead of changing its speed to produce much larger torques [4]; positioning the Cubli explicitly as a *reaction-wheel*, not CMG, demonstrator keeps the application claim precise.

Framed as an ADCS problem, the Cubli’s core behaviors map directly onto a spacecraft’s day-to-day pointing task:

- **Edge / corner balance** → regulation to a commanded reference attitude and disturbance rejection — holding a pointing setpoint against perturbations.
- **Controlled rotation** → a slew maneuver: reference tracking to a new commanded attitude, exactly as a satellite re-aims a payload or antenna.
- **Jump / walk** → momentum-exchange hopping mobility, flight-proven on small bodies where wheeled traction is impossible. JAXA’s Hayabusa2 mission deployed the MINERVA-II1 rovers [5] and the MASCOT lander [6] on asteroid Ryugu (2018), both of which move by spinning up and braking an internal flywheel or eccentric mass to hop across the surface, with MASCOT additionally self-righting the same way. Walking, as a sequence of

controlled, directional hops, is a genuine step beyond a single ballistic hop and is an active research frontier (e.g. ETH's SpaceHopper [7]).

This gives the project two complementary application narratives rather than one: it is simultaneously a low-cost, hardware-in-the-loop testbed for standard spacecraft *attitude determination and control* (ADCS), and a ground analog for the momentum-exchange *surface mobility* now used on small-body missions. Framed against a companion star-tracker project, which answers “*where am I pointing?*” (attitude determination), the Cubli answers “*drive me to that reference and hold it*” (attitude control) — the two halves of a complete ADCS loop.

Why the problem is still open. The nominal Cubli problem — balance on a corner, jump up — is solved, and was solved a decade ago on well-instrumented, precisely manufactured, generously funded hardware [1, 2, 3]. Reproducing that result is an engineering exercise, not a research contribution, and this document does not claim otherwise. The interesting questions begin immediately *after* that point, and they are not Cubli-specific: they are the questions a spacecraft attitude-control engineer faces, in a form small enough to be answered on a bench.

- **Control under simultaneous torque and momentum saturation.** A jump saturates the wheels by construction — the maneuver is defined by running them to their speed limit and dumping the momentum. Steering a nonlinear attitude system to a target while respecting a finite momentum envelope, and desaturating without losing the pointing objective, is still handled largely case-by-case with tuned heuristics rather than by a general method.
- **Recovery from large attitude excursions.** Linear controllers valid near upright say nothing useful once the body is far from equilibrium, which is precisely the regime a tumbling or mis-deployed spacecraft occupies. Nonlinear and predictive methods make claims there, and verifying those claims empirically requires a system in which a failed recovery costs a re-print rather than a mission.
- **Robustness to badly known parameters.** Real inertia tensors, friction coefficients and center-of-mass offsets are known to a few percent at best, and shift with assembly, thermal state and wear. Adaptive and robust designs are overwhelmingly validated against *simulated* parameter errors, which are drawn from the same distribution the designer assumed.
- **Whether high-performance methods survive commodity sensing.** Attitude-control techniques developed against clean measurements must eventually run on a MEMS IMU with drifting bias and vibration-corrupted accelerometers. Whether a given method degrades gracefully or collapses under that sensing is an empirical question, and it is settled by hardware rather than by analysis.

Why a Cubli, and why built from scrap. The Cubli is the cheapest honest test of the four questions above. Simulation cannot answer them, because the effects that decide the outcome are exactly the ones a model omits: unmodeled friction, frame resonance excited by the wheels, and the interaction between wheel imbalance and estimator bias. These are what separate a controller that works in simulation from one that works on hardware. At the other extreme, a full-scale testbed — air bearing, vacuum chamber, flight-grade wheels — costs orders of magnitude more and, decisively, *cannot be crashed repeatedly*. The scrap-built cube occupies the useful middle: physical enough to be honest, cheap enough to be abused. The choice to build it from commodity and salvaged parts is therefore methodological rather than merely budgetary, on four counts:

- **Cheap failure means the aggressive experiments actually get run.** A controller that must never fail cannot be tested at its limits, and a testbed too valuable to break silently biases the experimental program toward maneuvers already known to work.

- **Cheap parts are a stronger test than good parts.** Loose tolerances, an imperfectly balanced wheel and a drifting IMU are not defects to be apologized for — they *are* the parameter uncertainty and disturbance content that robust and adaptive methods claim to tolerate. A demonstration on deliberately imperfect hardware therefore carries evidence that a demonstration on precision hardware does not.
- **Constraints force honest sizing.** With a fixed motor, a fixed print volume and a fixed budget, a marginal design cannot be rescued by specifying a larger component. This is the entire justification for the bolt-in-pocket ballast scheme of Section 5.3.4: it buys inertia at the outer radius using standard M6 steel bolts and no machined part, and whether it closes is a calculation that must be completed *before* printing, not discovered afterwards.
- **A commodity bill of materials is reproducible.** Every part is orderable or salvageable by any university group, so the result can be checked by others — which is the difference between a demonstration and a claim.

Mission context. The subsystem exercised here is the one a spacecraft uses for pointing: momentum-exchange attitude control under actuator saturation, with desaturation, on an underactuated body whose inertia is imperfectly known. In one respect the terrestrial case is strictly *harder* than the orbital one. In orbit, a marginal attitude controller degrades slowly and often invisibly — pointing error grows, wheels creep toward saturation, and the fault may take orbits to become obvious. Under gravity, the destabilizing torque never vanishes and never relents: an inadequate controller falls over immediately and unambiguously. The demonstration is, in that sense, self-evaluating, and this is a feature of the platform rather than a limitation of it.

1.3 Objectives

The project targets four behaviors, deliberately ordered as a difficulty ramp in which each milestone is a control precondition for the next:

1. **Edge balance** — stabilize the cube on a single edge, the least unstable equilibrium (contact is a line, one primary axis of instability).
2. **Corner balance** — stabilize the cube on a single vertex, the fully unstable equilibrium (point contact, all three axes coupled and unstable simultaneously).
3. **Controlled rotation** — execute a commanded slew between equilibria (e.g. edge to edge, or in-place reorientation), demonstrating reference tracking rather than pure regulation.
4. **Walking** — chain repeated, directional controlled falls between edges/corners into a net translation across a surface, re-stabilizing after each transition.

These map one-to-one onto the applications of Section 1.2: (1)–(2) demonstrate pointing regulation and disturbance rejection of increasing difficulty; (3) demonstrates slew/reference-tracking; (4) demonstrates momentum-exchange mobility of the kind flown on Hayabusa2. Milestones (1)–(3) are treated as committed deliverables for this design; walking (4) is scoped as the stretch objective, contingent on hardware confirmation and the jump-up feasibility check (angular momentum required to tip the cube vs. available motor/wheel torque), to be resolved once final components are known.

2 Project Organization

2.1 Team Organization

The project is developed by a multidisciplinary team, with responsibilities distributed across the main technical domains of the Cubli: mathematical modeling, control, mechanical/CAD design, and system integration. Table 1 summarizes each member's primary role, and the detailed responsibilities are outlined below.

Table 1: Team members, contact details and primary roles.

Member	Contact	Primary Role
Pablo Urioste Alarcón	pb.urioste4@gmail.com +34 654 491 174	System integration and lead: digital actuator and sensor coding, embedded software, and integration of the overall design.
Deyan Nikolaev Vlaev	vlaev.dido@gmail.com +359 877 082 499	Mathematical demonstration and derivation of the system dynamics.
Niccolo	—	Control system design and analysis.
Andrea Liang	andrea.liang.2006@gmail.com +31 06 5796 2976	Support on the control problem and structural/mechanical integration.
Suvanna Viriyanti Wu	Viriyanti.wu@gmail.com +39 329 074 7550	CAD modeling and mechanical design.
Athanasia Nikolova	—	CAD modeling, prototype building/assembly, and product presentation (commercial pitch).

Responsibilities.

- **Pablo Urioste Alarcón** — System integration and overall coordination. Handles the coding of the digital actuators and sensors, the embedded software stack, and the integration of the mechanical, electronic and control subsystems into a working prototype.
- **Deyan Nikolaev Vlaev** — Mathematical foundation. Responsible for the analytical demonstration and derivation of the equations of motion and the dynamic model of the Cubli.
- **Niccolo** — Control. Leads the design, tuning and analysis of the control strategy used to balance and maneuver the Cubli.
- **Andrea Liang** — Control and mechanics support. Assists on the control problem and contributes to the structural and mechanical integration of the system.
- **Suvanna Viriyanti Wu** — CAD and mechanical design. Develops the 3D models and mechanical drawings of the structure and reaction-wheel assembly.
- **Athanasia Nikolova** — CAD, prototype building and commercialization. Contributes to the 3D modeling, leads the physical building and assembly of the prototype, and is responsible for presenting and pitching the product.

2.2 Work Breakdown Structure

This plan runs from project start to the final presentation on 20 August 2026. The documentation stream is front-loaded and closes with the Technical Design Document deliverable on 3 August,

while the control and hardware streams continue through the build, balancing and jump-up demonstrations up to the presentation. The work is grouped into three streams — investigation, requirements and documentation; control and software; and hardware design and build — led by the owners assigned in Table 1. The investigation is kept to the essentials needed to substantiate the design: a focused literature review, the dynamics derivation, and the wheel sizing that follows from it. The table below breaks the three streams into individual tasks, each with its scope, owner and dates.

ID	Task	Scope	Owner	Dates
Investigation, Requirements & Documentation (to the 3 Aug TDD deliverable)				
A1	Literature & state of the art	Review the ETH Cubli, the one-wheel Cubli, CubeSat ADCS practice and the hopping-mission precedents	Suvanna (+Dejan)	23–25 Jul
A2	Problem statement & framing	Lock the ADCS application narrative that the rest of the document refers back to	Dejan	23–24 Jul
A3	Scope & requirements	Define scope and build the requirements table (functional, performance, design constraint) with verification methods	Dejan (+Pablo)	24–25 Jul
A4	Milestones & roles	Prototype staging, binary gate criteria, and a one-page RACI of owners	Dejan	23–25 Jul
A5	Risk register	Trigger, impact, owner and mitigation for the leading risks	All (+Pablo)	28–29 Jul
A6	1-DoF equations of motion	Lagrangian derivation of the reaction-wheel pendulum with friction terms	Dejan (+Andrea)	24–25 Jul
A7	3D equations of motion	Coupled three-wheel dynamics for the edge and corner pivots	Dejan (+Andrea)	27–28 Jul
A8	Linearisation & controllability	Equilibria, linear models, controllability/observability checks	Dejan (+Andrea)	29 Jul
A9	Jump-up sizing & budget	Energy-method wheel sizing and the parametric mass/inertia/CoM budget	Dejan (+Andrea)	30 Jul–1 Aug
A10	TDD draft (all sections)	Every member drafts their section; placeholder figures acceptable	All	25–26 Jul
A11	TDD deepen + S1 data	Full modelling and control sections; insert the rig evidence and measured parameters	All	1–2 Aug
A12	TDD assemble & submit	Consistency pass, figures, references, submission of the deliverable	All (Dejan integrates)	3 Aug
Control & Software				
B1	Sim environment + plant	Simulation setup and the 1-DoF plant, validated by energy conservation	Andrea	23–24 Jul
B2	LQR design (1-DoF, sim)	Tune the regulator and sweep robustness to inertia error, delay and saturation	Andrea (+Dejan)	26–27 Jul
B3	State estimator	Complementary tilt filter with injected noise and bias in simulation	Nicc	27–28 Jul
B4	Firmware & comms	Firmware architecture, loop rates, and the CAN-FD command/reply layer	Nicc (+Pablo)	24–28 Jul
B5	Sensor drivers + cal	IMU and encoder drivers, gyro/accel calibration and the lever-arm correction	Nicc (+Andrea)	28–30 Jul
B6	Safety & mode logic	Watchdog, tilt-limit disarm, wheel-speed cap and the mode state machine	Nicc (+Andrea)	30–31 Jul

ID	Task	Scope	Owner	Dates
B7	S1 bring-up & balance	Port estimator and LQR to the rig and tune to a stable balance	Nicc (+Andrea)	31 Jul–2 Aug
B8	System identification	Measure K_t , wheel inertia, friction and loop latency on the rig	Andrea (+Nicc)	1–2 Aug
B9	3D model + LQR (sim)	3D plant and the edge/corner regulators with the CAD inertia tensor	Andrea (+Dejan)	3–6 Aug
B10	EKF / MEKF estimator	3D estimator with gyro-bias states; characterise the yaw drift	Nicc (+Andrea)	6–8 Aug
B11	Edge balance (S2a)	Retune the edge controller on the assembled cube; log disturbance recovery	Andrea (+Nicc)	10–13 Aug
B12	Corner balance + slew (S2b/c)	Coupled three-axis corner balance and a commanded slew; report ADCS metrics	Andrea (+Nicc)	14–16 Aug
B13	Jump-up (S3)	Spin-up profile and brake release for a face-to-edge jump	Nicc (+Andrea)	17–18 Aug

Hardware Design & Build

C1	Envelope & layout study	Packaging of wheels, drivers, battery and electronics with a central CoM	Neisa (+Suvanna)	23–24 Jul
C2	Reaction-wheel design	Wheel to the inertia target: radius, rim, hub bolting to the motor bell	Suvanna (+Neisa)	24–26 Jul
C3	Mounts & brackets	Motor mount, ring-magnet carrier and adjustable encoder bracket	Neisa, Suvanna	27–29 Jul
C4	1-DoF rig CAD + jigs	Single-axis rig and the balancing/assembly tooling	Suvanna (+Neisa)	25–28 Jul
C5	Mass properties & freeze	Export the inertia tensor/CoM and release the manufacturing package	Neisa (+Suvanna)	2–3 Aug
C6	Procurement order	Order the long-lead items (rims, frame stock, connectors, spares)	Pablo (+Neisa)	23–24 Jul
C7	Bench PSU & bring-up	Current-limited supply, safety kit, and single-axis electrical bring-up	Pablo (+Nicc)	25–27 Jul
C8	Fabricate wheels ($\times 4$)	Cut/receive rims, print hubs, bond magnets — three plus one rig spare	Neisa, Suvanna	27–30 Jul
C9	Balance & spin-test	Static-balance and containment spin-test every wheel	Suvanna (+Neisa)	30–31 Jul
C10	Assemble 1-DoF rig	Motor, balanced wheel, encoder gap, IMU and hard stops	Pablo (+Neisa)	31 Jul–1 Aug
C11	Print structure + frame	Structural print set and the fabricated, orthogonal frame	Neisa, Suvanna	3–6 Aug
C12	Perfboard + harness	Power/signal board and the full power and CAN harness	Pablo	3–6 Aug
C13	Cube integration + power-on	Assemble the three axes, then bench and battery smoke test	Pablo (+Neisa)	7–9 Aug
C14	Brake mechanism build	Fabricate and fit the servo brake; bench-test the deceleration	Suvanna (+Pablo)	10–13 Aug

Electrical & Electronics (electronics design: Nicc; wiring, testing and bring-up: Pablo)

E1	Bench power, e-stop & triage	Current-limited bench supply, e-stop and inline fuse; incoming test of all three actuation sets against the 194 m Ω phase figure, with CAN IDs assigned	Pablo	28–29 Jul
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ID	Task	Scope	Owner	Dates
E2	Driver, phase wiring & 5 V rail	Motor phase wiring and FOC calibration of the first set; LM2596 rail trimmed to 5.00 V and characterised under load to 1.5 A	Pablo (+Nicc)	29–30 Jul
E3	MA600 encoder + rig CAN bus	Magnet bonding, air gap ≤ 0.5 mm and bias trim; 2-node CAN-FD bus with split termination at both physical ends	Pablo	30–31 Jul
E4	Rig parameters + board outline	Measurement setup for K_t , wheel inertia, friction and loop latency; driver placement, CAN routing and the perf-board floorplan issued as a frozen outline	Nicc (+Pablo)	31 Jul–1 Aug
E5	1-DoF closed-loop bring-up	Teensy, IMU and CAN-FD closed loop on the rig; disarm path and watchdog verified before any balance attempt	Pablo (+Nicc)	1–2 Aug
E6	Perfboard + battery power chain	Populate the distribution perfboard (two ground domains, one star point, split 5 V rail); XT90 chain with anti-spark, fuse and ≥ 50 V bulk capacitance; replicate the remaining actuation sets	Nicc (+Pablo)	3–4 Aug
E7	4-node bus + cube harness	Bus scaled to four nodes with stubs < 10 cm and an error soak at 5 Mbit/s; full power and signal harness fabricated and routed; Wi-Fi telemetry for untethered operation	Pablo	5–7 Aug
E8	Cube power-on, 48 A + thermal	First battery power-on and the three-axis 48 A transient; driver thermals in the enclosed volume; all axes addressable	Pablo (+Nicc)	7–9 Aug
E9	Brake servo rail + measurement	Servo on an isolated 5 V rail with PWM drive and flyback; rail stability under stall, brake closure time and spin-down brake torque measured	Pablo (+Nicc)	10–18 Aug
Deliverables				
D1	Results, demo & final presentation	Slow-motion capture, results plots, deck, and the final presentation	All (+Suvanna edits)	17–20 Aug

2.3 Schedule and Contingency Plan

The full-page schedule in Figure 1 places every task of Section 2.2 on the 23 July–20 August 2026 calendar, and Table 3 states the gate that closes each phase. The documentation stream is deliberately completed early, at the TDD deliverable on 3 August, so the remaining weeks are free for the build, balancing and jump-up work that culminates in the final presentation. The critical path runs through the dynamics investigation and wheel sizing, which feed both the control and the hardware streams; the single-axis (S1) rig is the protected item, since its balance result is the experimental evidence in the document. If any modeling or rig task slips, the TDD is still submitted on schedule with simulation-only evidence and the affected result carried into the later build weeks.

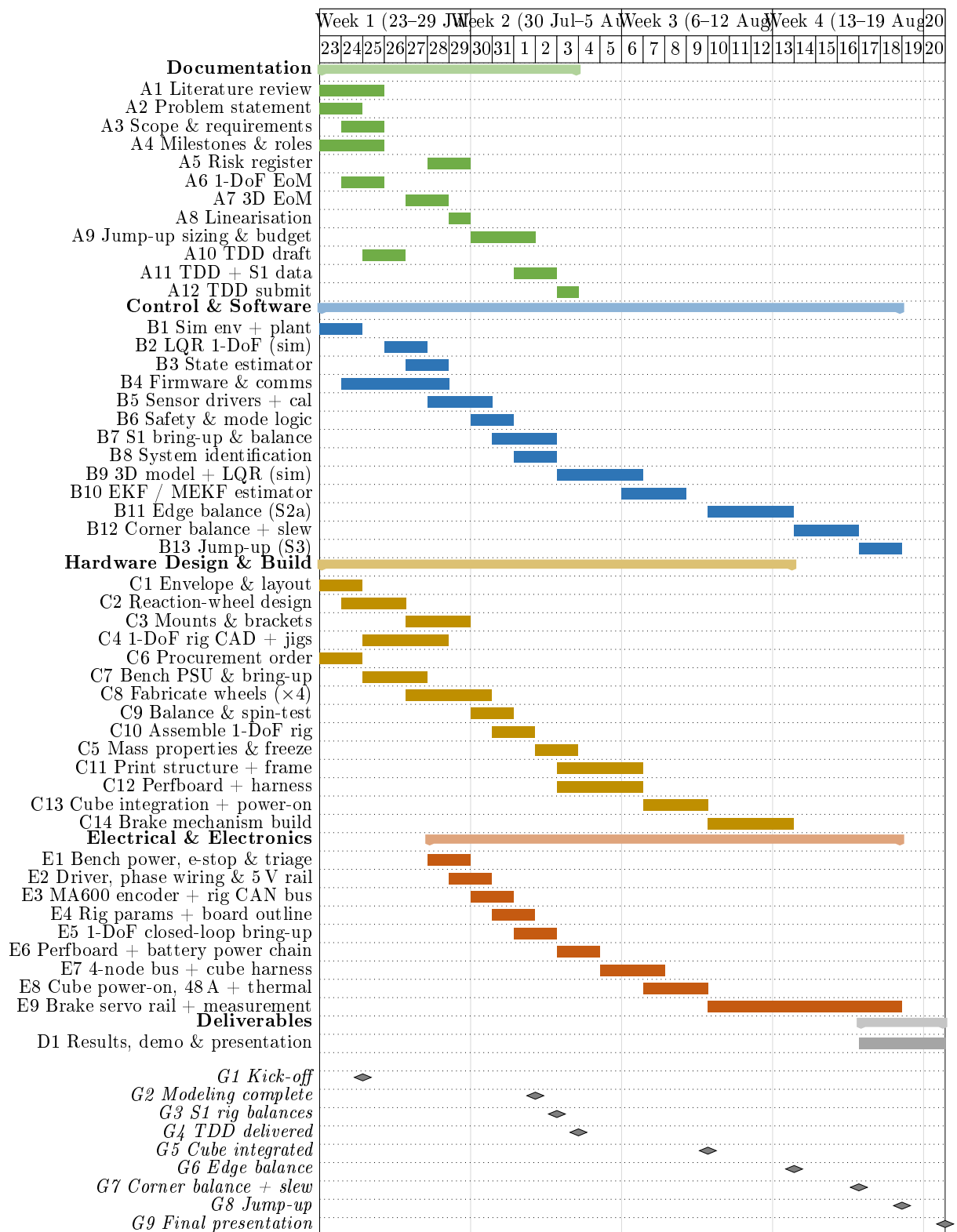


Figure 1: Detailed project schedule, 23 July–20 August 2026. The documentation stream (green) closes at the TDD deliverable on 3 August; control (blue), hardware (gold) and electrical (orange) continue through the build and demonstrations to the final presentation on 20 August. The electrical stream covers the design, wiring and testing of the 1-DoF rig (E1–E5, to G3), its extension to the 3D cube (E6–E8, to G5) and the brake actuation electrical (E9, to G8); each bar summarises several tasks of the detailed electrical schedule. Grey diamonds: milestone gates.

Table 3: Milestone gates, from kick-off to the final presentation.

Gate	Pass criterion	Date
G1	Long-lead order placed; requirements and plan locked	24 Jul
G2	Dynamics, control design and wheel sizing complete	1 Aug
G3	1-DoF rig balances ≥ 60 s; plant parameters measured	2 Aug
G4	TDD delivered (documentation deliverable)	3 Aug
G5	Cube integrated and powered; all axes addressable	9 Aug
G6	Edge balance (S2a) for ≥ 60 s with disturbance recovery	13 Aug
G7	Corner balance and commanded slew (S2b/c)	16 Aug
G8	Face-to-edge jump-up (S3) captured on video	18 Aug
G9	Final presentation and demonstration	20 Aug

3 System Analysis & Requirements

3.1 Scope of Work & Use Cases

3.2 Actuation Trade-off — Why Reaction Wheels

3.3 Requirements & Objectives Traceability

Each project objective (Section 1.3) imposes a required capability, which in turn is confirmed by a specific test in the validation plan (Section 6). The mapping below keeps every requirement traceable from motivation to verification.

Table 4: Objective-to-requirement-to-verification traceability.

Objective	Required capability	Verified by
Edge balance	Regulation about a line-contact equilibrium	Sec. 6
Corner balance	Regulation about a point-contact equilibrium, all axes	Sec. 6
Controlled rotation	Reference tracking / slew between equilibria	Sec. 6
Walking (stretch)	Chained jump-up + re-stabilization	Sec. 6

3.4 Coordinate Frames & Conventions

4 Dynamic Modeling & Control

The full Cubli is underactuated and simultaneously unstable about all three axes when balanced on a corner (Section 1.3). The modeling, identification, and control workflow is therefore first validated on a two-dimensional (2D) single-face prototype with one rotational degree of freedom, before being generalized to the 3D cube in Section 4.5.

4.1 2D Reaction-Wheel Pendulum

The 2D prototype is a 3D-printed plate sized to one cube face, with a motor-driven momentum wheel at its center and a braking mechanism at one corner. The plate pivots on a bearing at that corner, constraining motion to a single rotational degree of freedom in a vertical plane.

4.1.1 Equations of Motion

The nonlinear equations of motion are

$$\ddot{\theta}_b = \frac{(m_b l_b + m_w l) g \sin \theta_b - T_m - C_b \dot{\theta}_b + C_w \dot{\theta}_w}{I_b + m_w l^2}, \quad (1)$$

$$\begin{aligned} \ddot{\theta}_w = & \frac{(I_b + I_w + m_w l^2)(T_m - C_w \dot{\theta}_w)}{I_w(I_b + m_w l^2)} \\ & - \frac{(m_b l_b + m_w l) g \sin \theta_b - C_b \dot{\theta}_b}{I_b + m_w l^2}, \end{aligned} \quad (2)$$

with motor current-torque relation $T_m = K_m u$. Symbols are defined in Table 5 and used throughout this section.

Table 5: 2D pendulum model variables.

Symbol	Description	Unit
θ_b	Body tilt angle	rad
θ_w	Wheel rotation relative to body	rad
m_b, m_w	Body, wheel mass	kg
I_b, I_w	Body (about pivot), wheel (about motor axis) inertia	kg m ²
l	Motor axis to pivot distance	m
l_b	Body CoM to pivot distance	m
g	Gravitational acceleration	m/s ²
T_m	Motor torque	N m
C_b, C_w	Body, wheel damping coefficient	kg m ² /s
K_m	Motor torque constant	N m/A
u	Motor current input	A

4.1.2 Equilibria & Linearization

The relevant equilibrium is the upright position $(\theta_b, \dot{\theta}_b, \dot{\theta}_w) = (0, 0, 0)$; its linearization is presented with the control design in Section 4.4.

4.2 Parameter Identification & the 2D Testbed

CAD mass-properties simulation, accounting for print material, infill, and mounted hardware (driver, encoder, MCU, gyroscope/accelerometer, wiring), gives m_b, m_w, I_b, I_w directly, while l and l_b follow from the plate's design geometry. The damping coefficients C_b and C_w cannot be obtained from CAD, since they depend on bearing friction, wiring drag, and motor cogging; nominal values will be estimated for the initial control design, with dedicated testing planned once the prototype is fabricated.

Table 6: Parameter determination method (symbols per Table 5).

Symbol	Determination method
l	Calculated during design
l_b	Calculated during design
m_b	CAD mass properties
m_w	CAD mass properties
I_b	CAD mass properties
I_w	CAD mass properties
C_b	Estimated (future testing planned with prototype)
C_w	Estimated (future testing planned with prototype)
K_m	Motor datasheet

4.3 Jump-Up Feasibility Analysis

Given CAD-derived $I_b, I_w, m_b, l_b, m_w, l$, the wheel speed ω_w required for jump-up is estimated before fabrication, assuming a perfectly inelastic wheel-body collision. Conservation of angular momentum at impact and of energy from impact to edge-standing yield

$$\omega_w^2 = (2 - \sqrt{2}) \frac{I_w + I_b + m_w l^2}{I_w^2} (m_b l_b + m_w l) g. \quad (3)$$

This value is checked against the motor's achievable speed at the available bus voltage and used to size wheel inertia (radius, rim mass) within the motor's operating range without excessively reducing the balancing recovery angle. Post-fabrication, ω_w is refined experimentally to correct for deviations from the ideal collision assumption.

The braking mechanism uses a servo-driven barrier that strikes a bolt head on the wheel to stop it abruptly. Servo response time bounds the validity of the “instantaneous stop” assumption; a slower stop yields a lower effective post-impact body velocity than predicted, compensated through the same tuning process.

4.3.1 Torque-Limited Wheel-Inertia Target

The relation above assumes an instantaneous, lossless momentum transfer. For preliminary sizing, the achievable motor speed is treated as fixed and the required per-wheel inertia is derived from a momentum budget that additionally accounts for finite brake torque. This gives an ideal per-wheel angular momentum

$$h_{w,\text{ideal}} = \sqrt{\eta} \frac{\sqrt{2g\lambda\kappa}}{n} ML^{3/2}, \quad (4)$$

where M, L, g are cube mass, edge length, and gravity, η is an energy margin, and κ, λ, n are dimensionless geometric factors set by the contact case (Table 7).

The corner case is more demanding and governs the design.

Table 7: Contact-case geometric factors.

Case	κ (pivot inertia)	λ (CoM rise)	n (momentum transfer)
Edge	2/3	$1/\sqrt{2} - 1/2$	1
Corner (governs design)	11/12	$\sqrt{3}/2 - 1/2$	$\sqrt{2}$

A finite brake torque τ_b cannot deliver the impulse instantaneously and must exceed the gravitational tip-over torque $\tau_g = MgL/2$. This loss is captured by a brake-amplification factor

$$\beta = \frac{\tau_b}{\tau_b - \tau_g}, \quad h_w = \beta h_{w,\text{ideal}}, \quad I_{w,\text{target}} = \frac{h_w}{\omega_{\max}}, \quad (5)$$

where $\omega_{\max}$ is the maximum practical wheel speed. Equations (4) and (5) are the sizing relations used at step 4 of the closure procedure of Section 5.2: given a candidate cube mass and edge length they return the inertia the wheel of Section 5.3.4 must meet. Their inputs are recorded in Table 8 and are not fixed here— M and L are outputs of the packaging closure, not assumptions preceding it.

Two features of these relations govern how the iteration behaves. First, $h_w \propto ML^{3/2}$, so growing the cube to accommodate a larger wheel also raises the requirement; convergence depends on the achievable inertia growing faster ($\sim D_w^2$) than the requirement does. Second, β diverges as $\tau_b \rightarrow \tau_g$: a brake whose torque only marginally exceeds the gravitational tip-over torque cannot deliver the required impulse at any wheel speed. The brake torque must therefore be checked against $\tau_g = MgL/2$ on every iteration, and a comfortable ratio τ_b/τ_g maintained rather than a merely positive margin.

Table 8: Torque-limited wheel-inertia target at the current design point. Inputs are fixed by the closure procedure of Section 5.2; the table is recomputed on each iteration. M is provisional (Section 5.2) and τ_b is not yet substantiated (Section 4.3.2); both are flagged where they are used.

Symbol	Description	Value
M	Cube mass (provisional)	1.40 kg
L	Cube edge length	180 mm
η	Energy margin	1.05
τ_b	Brake torque per wheel (unverified)	5.0 N m
$\tau_g = MgL/2$	Gravitational tip-over floor	1.236 N m
β	Brake-amplification factor	1.328
$\omega_{\max}$	Max wheel speed (design)	6000 rpm (628 rad/s)
h_w	Required angular momentum	0.2640 kg m ² /s
$I_{w,\text{target}}$	Target wheel inertia	4.202×10^{-4} kg m²

Table 14 gives both contact cases at this design point. The corner case governs, as anticipated: it demands 10.2% more inertia than the edge case at the same wheel speed.

Choice of $\omega_{\max}$. The maximum wheel speed is a design choice rather than a datasheet figure, and it enters the sizing directly through $I_{w,\text{target}} = h_w/\omega_{\max}$: a higher assumed speed buys a proportionally smaller wheel. It is therefore the natural relief valve when the envelope caps the wheel below the requirement (step 5, Table 13), but it must be set from what the drive can sustain in service, not from the no-load figure $K_v V_{\text{bus}}$. Three effects separate the two: winding

resistance, which costs speed under the torque required to spin the wheel up; the reduced phase voltage available under sinusoidal field-oriented modulation; and the fall of pack voltage through the discharge curve, which must be taken at its lower end if the maneuver is to remain available on a partly-depleted pack rather than only on a fresh one. The design value is set below the ceiling those effects imply, leaving margin for motor tolerance, bearing and aerodynamic drag, and the residual uncertainty in M . The resulting back-EMF should remain a comfortable fraction of the usable bus voltage, and the electrical frequency $\omega_{\max}p/2\pi$ at p pole pairs must sit inside the driver's commutation capability.

Applying this to the selected motor and pack fixes the design speed at **6000 rpm** (628 rad/s). The derivation is Table 9. At 380 rpm/V the MN4006 reaches 8436 rpm unloaded on a nominal 22.2 V pack — an upper bound with no load and no losses, not an operating point. Subtracting the winding drop across $2R_{\text{phase}} = 0.388 \Omega$ at a representative spin-up current, then derating by a factor of 0.90 for sinusoidal modulation against a trapezoidally-rated K_v , and finally taking the pack at its end-of-charge voltage, leaves a realistic ceiling near 6900 rpm. The design value sits 13 % below even that.

Table 9: Wheel-speed ceiling by pack state, after winding drop at a representative 2 A spin-up current and a 0.90 modulation derate. The design value is set below the sagged-pack figure.

Pack state	Bus voltage	Ideal $K_v V$	Achievable
Charged	25.2 V	9576 rpm	8353 rpm
Nominal	22.2 V	8436 rpm	7327 rpm
Sagged	21.0 V	7980 rpm	6917 rpm
Design value	—	—	6000 rpm

The conservatism is deliberate and is bought rather than conceded. At 6000 rpm the back-EMF is 15.79 V, only 71 % of the nominal bus, so the wheel reaches its commanded speed across essentially the whole discharge curve rather than only on a freshly charged pack — the relevant failure mode being a machine that works on the first attempt after charging and degrades through a demonstration session. Every derived quantity also becomes less demanding, since all scale with $\omega_{\max}^2$: stored energy is 89 J per wheel rather than 133 J at 7327 rpm, and the ballast retention load falls in the same proportion (Section 5.3.4). The commutation check is not binding: at 12 pole pairs, 6000 rpm is an electrical frequency of 1200 Hz, well inside the driver's capability, so the constraint on top speed is the battery and not the driver.

The cost is a larger wheel. Since $I_{w,\text{target}} \propto 1/\omega_{\max}$, designing at 6000 rpm rather than at a top-of-range assumption demands roughly 40 % more inertia. The selected wheel of Section 5.3.4 absorbs this, which is what makes the margin affordable.

4.3.2 Brake Torque and Its Provenance

The brake torque τ_b enters the sizing through β alone, but it does so steeply, and the value used in Table 8 is the least substantiated input in the chain. It is recorded here as an open item rather than presented as a result.

The brake is a servo-driven barrier struck by a ballast bolt head, so τ_b is *not* the servo's output torque: the servo only holds the barrier in place while the wheel's own momentum does the work against it. The figure is therefore a structural and impulse quantity to be measured on the rig, and the supplied micro servo's rated output — some seventeen times smaller — is not the relevant comparison. What the servo must supply is enough holding torque and fast enough travel that the barrier is in position before the bolt arrives.

The sensitivity is the reason this matters. Because $\beta = \tau_b/(\tau_b - \tau_g)$ diverges as $\tau_b \rightarrow \tau_g$, an overestimate of τ_b propagates directly into an undersized wheel (Table 10). At the assumed 5.0 N m the penalty is a modest 33 %; at 2.5 N m it is 105 %, which would raise the inertia target above what the selected wheel can reach and force a redesign. Measuring τ_b on the 1-DoF rig is consequently a precondition for freezing the wheel, not a confirmation exercise afterwards.

Table 10: Sensitivity of the inertia target to the assumed brake torque, at $M = 1.40$ kg, $L = 180$ mm ($\tau_g = 1.236$ N m) and $\omega_{\max} = 6000$ rpm. The assumed value is not yet measured.

τ_b (N m)	β	$I_{w,\text{target}}$ (10^{-4} kg m ²)
2.0	2.62	8.28
2.5	1.98	6.26
3.0	1.70	5.38
4.0	1.45	4.58
5.0 (assumed)	1.33	4.20
7.0	1.21	3.83

4.4 Control Design on the 2D Model

4.4.1 LQR Baseline Control

Linearizing about the upright equilibrium $(\theta_b, \dot{\theta}_b, \dot{\theta}_w) = (0, 0, 0)$ gives

$$\dot{x}(t) = Ax(t) + Bu(t), \quad x := (\theta_b, \dot{\theta}_b, \dot{\theta}_w), \quad (6)$$

with A, B built from the parameters in Table 5. Zero-order-hold discretization at a rate set by the open-loop unstable pole, with a safety margin, gives $x[k+1] = A_d x[k] + B_d u[k]$. A discrete LQR gain $K_d = (K_{d1}, K_{d2}, K_{d3})$ minimizes

$$J(u) = \sum_k x^T[k] Q x[k] + u^T[k] R u[k], \quad (7)$$

giving $u[k] = -K_d(\hat{\theta}_b[k], \hat{\theta}_b[k], \hat{\theta}_w[k])$.

The tilt estimate $\hat{\theta}_b$ is obtained from a gyroscope and accelerometer mounted at, or as close as mechanically feasible to, the pivot point rather than at the plate's center. This placement is necessary because an accelerometer at radius r from the pivot measures gravity plus the tangential ($r\ddot{\theta}_b$) and centripetal ($r\dot{\theta}_b^2$) acceleration components induced by the plate's rotation, which corrupt the naive estimate $\tan^{-1}(-a_x/a_y)$ during fast motion (e.g., jump-up or aggressive recovery); placing the sensor at $r \approx 0$ makes both terms negligible. Tilt is estimated with a complementary filter combining gyro integration (drift-prone, unaffected by the terms above) with accelerometer inclination (drift-free, but unreliable during shocks):

$$\hat{\theta}_b[k] = (1 - \gamma)(\hat{\theta}_b[k-1] + \dot{\theta}_{b,\text{gyro}}[k] \Delta t) + \gamma \theta_{b,\text{accel}}[k], \quad (8)$$

where $\theta_{b,\text{accel}}[k] = \tan^{-1}(-a_x[k]/a_y[k])$ is the instantaneous accelerometer-only estimate and $\gamma \in (0, 1)$ is a small weight favoring the gyro term between corrections.

A constant tilt-sensor offset d , including residual bias from imperfect pivot placement or sensor misalignment, would otherwise appear as a nonzero steady-state wheel speed rather than a corrected angle. A low-pass filter state

$$x_f[k+1] = (1 - \alpha)x_f[k] + \alpha\hat{\theta}_b[k] \quad (9)$$

is subtracted from the tilt feedback,

$$u[k] = -K_d(\hat{\theta}_b[k] - x_f[k], \hat{\theta}_b[k], \hat{\theta}_w[k]), \quad (10)$$

so d is absorbed by x_f at steady state, requiring no separate calibration stage. This fixed-gain design is the baseline balancing controller; Table 11 summarizes the control variables.

Table 11: Control-design variables (2D and corner-balancing 3D model).

Symbol	Description
x	State vector
A, B	Continuous-time state/input matrices (linearized model)
A_d, B_d	Discrete-time (ZOH) counterparts of A, B
K_d	LQR feedback gain
Q, R	LQR state/input weighting matrices
$J(u)$	LQR cost function
$\hat{\theta}_b, \dot{\hat{\theta}}_b$	Estimated tilt angle, rate (gyroscope/accelerometer)
$\hat{\theta}_w$	Estimated wheel rate (encoder)
a_x, a_y	Raw accelerometer readings (tangential, radial axes)
$\theta_{b,\text{accel}}$	Instantaneous accelerometer-only tilt estimate
$\dot{\theta}_{b,\text{gyro}}$	Raw gyro rate measurement
γ	Complementary-filter weight (accelerometer term)
Δt	Control/estimation loop sample period
x_f, α	Offset-correction filter state, filter coefficient
d	Constant tilt-sensor offset
θ_{th}	Tilt threshold, jump-up $\rightarrow$ balancing switch
$K_d^{(i)}$	Scheduled gain at linearization point i (extension)

4.4.2 Gain Scheduling & Mode Transitions

The controller switches from the open-loop jump-up sequence to the LQR law once $|\hat{\theta}_b| < \theta_{\text{th}}$, with hysteresis around θ_{th} to prevent mode chattering. The baseline uses a single fixed gain K_d at the upright equilibrium. If recovery degrades near the edges of the operating range, where the small-angle linearization weakens, gain scheduling will be added: additional gains $K_d^{(i)}$ computed at several linearization points, with the controller interpolating between them based on $\hat{\theta}_b$, reusing the same discrete structure and offset filter.

4.5 Generalization to the 3D Cube

The dynamics, identification workflow, and control approach validated on the 2D model extend to the full three-dimensional cube, where three orthogonal reaction wheels must stabilize the body about its edge and corner equilibria.

4.5.1 Full-Body Equations of Motion

With $R \in SO(3)$ the cube's orientation and $\omega_b \in \mathbb{R}^3$ its body-frame angular velocity, the attitude kinematics are

$$\dot{R} = R[\omega_b]_{\times}, \quad (11)$$

where $[\cdot]_{\times}$ is the skew-symmetric cross-product matrix. The assembly's angular momentum about the pivot, expressed in the body frame, generalizes the single-axis momentum term of the 2D model:

$$H = I\omega_b + \sum_{i=1}^3 I_{w,i} \omega_{w,i} e_i. \quad (12)$$

Euler's equation gives

$$\dot{H} + \omega_b \times H = \tau_g - C_b \omega_b, \quad (13)$$

and each wheel obeys

$$I_{w,i}(\dot{\omega}_{b,i} + \dot{\omega}_{w,i}) = T_{m,i} - C_{w,i} \omega_{w,i}, \quad i = 1, 2, 3. \quad (14)$$

Table 12 defines all symbols.

Table 12: Full-body (3D) model variables.

Symbol	Description
R	Cube orientation relative to world frame
ω_b	Body angular velocity, body frame ($\in \mathbb{R}^3$)
$\omega_{b,i} = e_i^T \omega_b$	Body angular velocity about wheel axis i
$\omega_w = (\omega_{w,1}, \omega_{w,2}, \omega_{w,3})$	Wheel angular velocities relative to body
e_i	Spin axis of wheel i (face normal)
I	3×3 inertia tensor, body+wheels, about pivot
H	Total angular momentum about pivot
$\tau_g = m r_{cm} \times (R^T \hat{g})$	Gravity torque about pivot
m	Total assembly mass
r_{cm}	Body-frame vector, pivot to assembly CoM
$\hat{g}$	World vertical unit vector
C_b	3×3 body damping matrix
$I_{w,i}, C_{w,i}$	Inertia, damping of wheel i
$T_{m,i} = K_{m,i} u_i$	Motor torque, wheel i
$u_i, K_{m,i}$	Motor current, torque constant, wheel i

Restricting motion to a single vertical plane with two wheel torques set to zero recovers the 2D prototype model exactly, with $I_b, l_b, m_b, m_w, I_w, C_b, C_w$ the corresponding single-axis projections of $I, r_{cm}, m, I_{w,i}, C_{w,i}$.

4.5.2 Edge vs. Corner Equilibria & Linearization

Jump-up proceeds in two stages: Flat-to-Edge, where one wheel brakes, followed by Edge-to-Corner, where a second wheel brakes. **Edge balancing** constrains the body to a single axis, so the dynamics reduce to the 2D form, and the 2D LQR baseline and offset filter apply directly, with m_b, l_b, I_b replaced by their edge equivalents.

Corner balancing is genuinely 3D: all three wheels contribute across three rotational DOF. Linearizing about the corner-up equilibrium $(\theta, \omega_b, \omega_w) = (0, 0, 0)$, with $\theta \in \mathbb{R}^3$ the small orientation error, gives the same state-space form as the 2D case:

$$\dot{x}(t) = Ax(t) + Bu(t), \quad x := (\theta, \omega_b, \omega_w) \in \mathbb{R}^9, \quad (15)$$

with $A \in \mathbb{R}^{9 \times 9}$ and $B \in \mathbb{R}^{9 \times 3}$ built from I, r_{cm} , and the per-wheel parameters $I_{w,i}, C_{w,i}, K_{m,i}$ (Table 12). The LQR design generalizes directly: K_d becomes a 3×9 gain matrix, using the same

discretization, cost function, and offset filter as Table 11, with Q, R weighting all three tilt axes, body rates, and wheel speeds. The mode-transition logic gains a third mode, Edge-to-Corner jump-up, triggered once edge balancing is stable, followed by a second threshold switch into the corner LQR once estimated orientation nears corner-up.

5 Preliminary Hardware Design

This design is preliminary in a specific sense. The actuation, sensing and power components are selected (Appendix A) and the principal dimensions follow from the closure of Section 5.2: a 180 mm cube with three 130 mm ballasted reaction wheels at a 6000 rpm design speed. What remains open is the structure that carries them, and two inputs on which those dimensions depend — the total mass M and the brake torque τ_b — are not yet measured. The architecture is therefore kept parametric so that the closure can be re-run against measured values rather than re-derived (Section 5.3.1), and the two open inputs are flagged wherever they are used rather than presented as settled.

5.1 System Architecture

Figure 2 shows the closed control loop that ties the four subsystems together. The Teensy 4.1 runs estimation and control at 1 kHz and issues torque commands over CAN-FD to three moteus-n1 drivers, each closing a 30 kHz field-oriented current loop on its MN4006 motor. The resulting reaction torque acts on the cube body; the body’s attitude is sensed by the BMI270 IMU and the rotor angle by the MA600 absolute encoder, closing the loop back to the controller. The two loops run at different rates by design: the fast current loop is delegated entirely to the drivers, so the flight controller need only meet the 1 ms outer-loop deadline of Section 4.4.

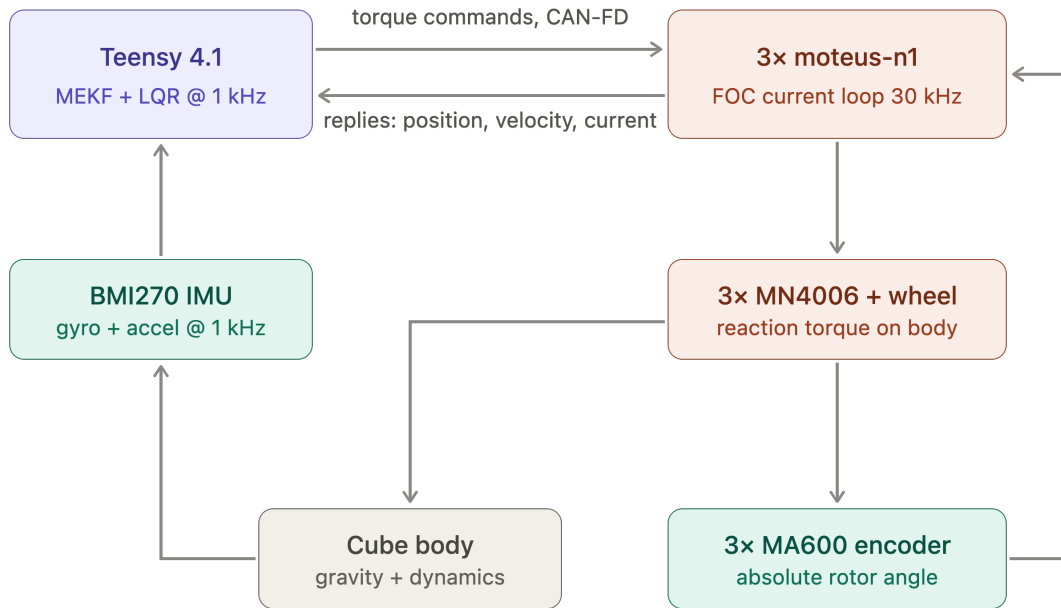


Figure 2: System architecture and control loop. Blue: processing; green: sensing; orange: actuation. Rates shown are the design targets of Section 4.

5.2 Sizing Constraints and Closure Procedure

The cube edge length L and the reaction-wheel inertia I_w are not independent design choices: they are the two ends of a single constraint chain that begins with what must physically be carried inside the cube. This section states that chain, the order in which it is closed, and the envelope it selects: $L = 180$ mm, evaluated in Section 5.2.1. The numbers follow from the procedure rather than preceding it, and the closure is re-run whenever an input moves.

The battery and the internal structure drive the envelope. The dominant volume claim inside the cube is not the reaction wheels but the power and support hardware. A LiPo pack sized for the jump-up duty cycle is a monolithic, incompressible block: unlike the electronics it cannot be redistributed into the corners or thinned, and its dimensions follow from the required capacity and discharge rating rather than from any packaging preference. Around it must sit an *inner structure*—a rigid subframe that locates the three orthogonal motors on mutually perpendicular axes through a common centre, carries the three motor drivers, the flight controller, the IMU and the brake servos, and reacts the motor mounting loads into the outer frame. The three wheel/motor modules must not intersect one another, the pack, or this subframe, and the pack should sit near the geometric centre so that it does not skew the cube’s centre of mass away from the body diagonal.

The minimum internal clear volume is therefore set by the pack, the subframe and the electronics stack together, and L is the smallest edge length that contains that volume plus the wall thickness and the corner and edge contact features. This is the first quantity to fix, and everything downstream depends on it.

The envelope caps the wheel, and the wheel must meet the momentum requirement.

Once L is set, the largest wheel outer diameter D_w that clears the subframe, the motor body and the adjacent orthogonal modules is a geometric consequence, and it caps the achievable inertia—not only through $I_w \sim m_w D_w^2/4$, but because any ballast necessarily sits inboard of the ring it displaces, so a wheel that is short of target at its maximum diameter cannot be rescued by adding mass. Against this stands the requirement from the jump-up analysis (Section 4.3.1),

$$I_{w,\text{target}} = \frac{h_w(M, L)}{\omega_{\max}}, \quad h_w \propto ML^{3/2},$$

which grows with both the cube mass and the edge length. Enlarging the cube to gain wheel diameter therefore also raises the requirement—but the achievable inertia grows faster than the requirement does (D_w^2 against $L^{3/2}$), so the loop does converge, and a wheel that is short at a given L can be closed by a modest increase in L .

Mass closes the loop. The cube mass M is not merely a logistics figure: it re-enters the dynamics through the gravitational tip-over torque $\tau_g = MgL/2$, which sets the floor the brake torque must exceed, and through $h_w \propto M$. Growing the envelope to fit a larger wheel adds frame and wheel mass, which raises $I_{w,\text{target}}$ again. Mass, volume and inertia are thus one coupled system, closed by the iteration of Table 13 rather than by any of the three in isolation.

Inertia requirement is corner-critical. The two contact cases are not equally demanding. The corner equilibrium requires a larger rise of the centre of mass and a less favourable momentum-transfer geometry than the edge equilibrium, so for a common maximum wheel speed $\omega_{\max}$ it demands the greater wheel inertia (Table 7). Step 4 above is therefore always evaluated for the corner case; a wheel that satisfies the corner requirement satisfies the edge requirement with margin. Table 14 records both once the procedure closes.

5.2.1 Envelope Selection: Why 180 mm

Running the procedure over candidate edge lengths selects $L = 180$ mm. Two independent arguments converge on it, and both are needed — the dynamic one alone would not settle the question.

The inertia budget. Table 15 evaluates step 4 against step 3 for each candidate. The requirement grows as $ML^{3/2}$ and is further penalised by β , which degrades with L ; against this, the

Table 13: Sizing closure procedure. Steps are executed in order; a failure at step 4 returns to step 3 or step 1 as indicated.

Step	Determines	Governing consideration
1	Battery pack geometry	Capacity and discharge rating for the jump-up duty cycle
2	Inner clear volume $V_{\min}$, hence minimum L	Pack + motor subframe + driver/controller stack, non-intersecting; pack near centre
3	Maximum wheel diameter D_w , hence achievable I_w	Clearance to subframe, motor body and the two orthogonal modules
4	Requirement $I_{w,\text{target}} = h_w(M, L)/\omega_{\max}$	Corner jump-up, Eqs. (4)–(5)
5	Accept or iterate	If $I_w < I_{w,\text{target}}$: raise L (step 2), raise $\omega_{\max}$, or trim M ; re-enter at step 3
6	Mass reconciliation	Updated M feeds back into step 4; iterate to a fixed point

Table 14: Required per-wheel inertia by contact case, at $M = 1.40$ kg, $L = 180$ mm and $\omega_{\max} = 6000$ rpm. The corner case governs by 10.2 %.

Contact case	h_w (kg m ² /s)	$I_{w,\text{target}}$ (10 ⁻⁴ kg m ²)
Edge	0.2395	3.812
Corner (critical)	0.2640	4.202

largest wheel the frame admits is roughly L less a 30–40 mm allowance for wall thickness, motor boss and contact features, and the inertia it supplies grows as the square of that diameter. At 150 mm the largest wheel that fits is about 120 mm in diameter and supplies 2.63×10^{-4} kg m² against a 3.03×10^{-4} kg m² requirement — 13 % short, and *not* recoverable by ballast, since ballast necessarily sits inboard of the ring it displaces. At 180 mm the requirement is 4.20×10^{-4} kg m² and a 130 mm wheel supplies 3.49×10^{-4} kg m² from its printed structure alone, closing the remainder with ballast and retaining trim authority in both directions.

Table 15: Envelope trade at $\omega_{\max} = 6000$ rpm, $M = 1.40$ kg. The wheel cap assumes a 30 mm frame allowance; bare-ring inertia is for the 20 mm $\times$ 12 mm ring section of Section 5.3.4.

L (mm)	β	$I_{w,\text{target}}$ (10 ⁻⁴)	Bare ring at cap (10 ⁻⁴)	Verdict
150	1.259	3.03	2.63	Short by 13 %
160	1.282	3.40	3.49	Closes with no trim margin
180	1.328	4.20	5.73	Clears; 130 mm wheel selected
200	1.379	5.11	8.78	Clears, but heavier and β worse

Packaging, which is independently decisive. The dynamic argument alone would leave 160 mm arguable. Packaging does not: the power and signal board is 150 mm $\times$ 90 mm, so it does not fit inside a 150 mm envelope at any wall thickness, and the interior must additionally carry three motors, three drivers, a 6S pack and the brake servos (Section 5.4.5). This is the constraint that removes the smaller options outright, and it is why step 2 of Table 13 precedes the wheel sizing rather than following it.

What the decision costs. A larger cube is not free, and the penalty should be stated rather than buried. Frame mass grows with L , and M is already the least comfortable number in the

budget (Table 16); β degrades from 1.259 at 150 mm to 1.328 at 180 mm, so the finite brake loses a larger fraction of its impulse to gravity; and both effects feed back into $I_{w,\text{target}}$. Note that β depends entirely on the unverified brake torque of Section 4.3.2, so this column of the trade is the least secure part of it. The design is accepted at 180 mm because the alternative does not close at all, not because the cost is negligible.

This makes the cube larger than the 150 mm ETH prototype referenced in Section 1.1, and the reason is material rather than dimensional: that design used a machined steel flywheel, where this one uses a printed ring with steel ballast at a substantially lower achievable density at radius. The lost density is compensated with radius, and radius is bought with envelope.

Mass budget. The allocation of Table 16 is the bookkeeping side of steps 5 and 6: it is populated as components are selected and re-totaled on each iteration, since the total is an input to $I_{w,\text{target}}$ and not merely an output of the design.

Table 16: Mass budget (per cube). Known figures are from the selected components; allowances are provisional. The total is an input to the inertia requirement, not only a result — see the reconciliation below.

Subsystem	Mass (g)	Basis
Battery pack	600	Selected 6S pack
Reaction wheels (3×)	468	Sec. 5.3.4, 156 g each
Motors (3×)	204	Datasheet, 68 g each
Drivers (3×)	44	Datasheet, 14.6 g each
<i>Subtotal, known</i>	<i>1316</i>	<i>Fixed by selection</i>
Inner structure / motor subframe	TBD	Pending CAD
Outer frame / contact features	TBD	Pending CAD
Electronics (MCU, board, IMU, servos)	TBD	Allowance, pending weigh-in
Wiring / fasteners / margin	TBD	Allowance
Design value M	1400	Used in Sec. 4.3.1

The mass budget does not currently close, and this is the leading design risk. The four selected components above total 1316 g, which is 94 % of the 1400 g figure that Section 4.3.1 uses as an input — with no frame, no electronics board, no wiring, no fasteners and no brake hardware yet counted. The remaining allowance of 84 g is not credible for those items.

This is precisely the fixed-point failure that Section 1.1 identifies as the characteristic way Cubli replications fail, and the honest statement is that the design is currently on the wrong side of it. The coupling makes it worse than a simple overrun: $h_w \propto M$, so a realistic M raises $I_{w,\text{target}}$, which demands a heavier wheel, which raises M again. Three responses are available, in order of preference:

1. **Reduce structural mass.** The frame and subframe are the only large uncommitted allocations, and they are the parts the CAD can still influence. This is the first recourse and it is why the mass roll-up is an output of the model (Section 5.3.1) rather than an annotation on it.
2. **Accept a higher M and re-close.** The wheel retains upward trim authority (Section 5.3.4), so a moderate increase can be absorbed by installing additional ballast rather than by re-designing. The margin available for this is quantified there and it is not large.

3. **Raise $\omega_{\max}$.** Since $I_{w,\text{target}} \propto 1/\omega_{\max}$, recovering the deliberate conservatism of Section 4.3.1 is the relief valve of last resort. It is listed last because it trades away the discharge-curve robustness that motivated the choice, and every load in the machine scales with $\omega_{\max}^2$.

The measured mass at first assembly is therefore a gating figure, not a reporting one: Appendix B.6 records it against the model, and the closure of Table 13 is re-run on that number before the wheels are considered frozen.

5.3 Mechanical Design and CAD Construction

This subsection states the mechanical concept and the CAD conventions the build follows. It is deliberately kept to the level of decisions and interfaces: the part-by-part model tree, the manufacturing drawings, the fabrication settings and the assembly sequence are recorded in Appendix B, so that a change to a bracket does not require an edit to the design rationale, and the rationale is not buried in geometry.

5.3.1 CAD Methodology and Model Conventions

The model is built top-down from the envelope closure of Section 5.2 rather than bottom-up from individual parts, so that the edge length L and the wheel diameter D_w propagate to every dependent feature when an iteration changes them. Three conventions make that propagation reliable, and they are stated here because they constrain how every part in Appendix B is modelled:

1. **A single skeleton drives the assembly.** L , D_w , the motor axis offsets and the wall thickness live in one parameter table referenced by every part; no downstream part carries a hard-coded dimension that duplicates one of them.
2. **The body frame is the CAD origin.** The assembly origin is the cube's geometric centre, with axes on the three motor spin axes, so that CAD-reported mass properties are directly the I and r_{cm} of Table 12 without a change of basis.
3. **Mass properties are an output of the model, not an annotation.** Every part carries its assigned material and print density, so the assembly mass roll-up is the quantity fed back into Table 16 at step 6 of Table 13.

Convention 3 is what makes the iteration auditable: the mass budget and the inertia requirement are read from the same model, so a design that closes on paper cannot silently fail to close in the CAD.

5.3.2 2D Prototype Build

The single-face rig is the first article built and the bench on which the parameters of Table 6 are measured. Its structure is a 3D-printed plate sized to one cube face, a central motor and wheel mount, a corner bearing defining the single rotational degree of freedom, and the servo brake. Detailed geometry and the drawing set are in Appendix B.3.

5.3.3 3D Cube Frame

The frame is conceived in two parts, reflecting the constraint chain of Section 5.2. An *inner structure* locates the three motors on mutually perpendicular axes through a common centre and carries the battery pack, the three drivers, the flight controller and the brake servos; it is the element that fixes the minimum internal clear volume and therefore the edge length L . An *outer frame* carries the edge and corner contact features, protects the wheels, and reacts the motor

mounting loads from the inner structure into the ground contact. Detailed geometry is pending CAD and follows the envelope closure.

The split is also the main mechanical interface in the machine, and it is managed as one: the inner structure is designed against a fixed bolt pattern and keep-out envelope published to the outer frame, so the two can be revised independently as long as that interface holds. The controlled interfaces are listed in Table 17; the parts realising them, with their drawings, are in Appendix B.4.

Table 17: Controlled mechanical interfaces. Each is frozen once agreed, so that the parts on either side can be revised independently.

Interface	Between	Controlled quantities
Motor mount	Motor $\leftrightarrow$ inner structure	Bolt circle, shaft height, concentricity to the wheel axis
Wheel hub	Wheel $\leftrightarrow$ motor rotor	Bore, bolt pattern, axial location, balance datum
Frame joint	Inner structure $\leftrightarrow$ outer frame	Bolt pattern, keep-out envelope, load path to the contact features
Electronics mount	Board $\leftrightarrow$ inner structure	Board outline, mounting-hole pattern, keep-out heights, connector exit directions (Sec. 5.4.5)
Brake mount	Servo $\leftrightarrow$ inner structure	Servo location, barrier throw, clearance to the ballast bolt heads
Battery retention	Pack $\leftrightarrow$ inner structure	Pack envelope, retention method, centring tolerance to the cube centre
Contact features	Outer frame $\leftrightarrow$ ground	Corner and edge geometry, material, replaceability

5.3.4 Reaction-Wheel Sizing

The wheel must supply the per-wheel inertia $I_{w,\text{target}}$ demanded by the corner jump-up (Section 4.3.1), at an outer diameter no greater than the envelope permits (step 3 of Table 13), while remaining light enough not to dominate the cube mass—since its own mass feeds back into $I_{w,\text{target}}$ through M . These three demands pull against each other, and the wheel architecture is chosen so that the design can be trimmed against them after fabrication rather than having to be right the first time.

Architecture: ballasted ring. Rather than adjusting inertia by scaling a solid disc, the wheel is built as a printed *ring plus spokes* whose inertia is then tuned by *ballast*: standard steel bolts and nuts seated in blind hexagonal pockets distributed around the ring. Each pocket both removes plastic and adds steel at the largest available radius, so the net inertia gain per pocket is large, and the target is reached by choosing how many bolts to install. This makes wheel inertia a discrete, reversible knob: the physical wheel can be re-tuned in either direction after fabrication to absorb the parameter uncertainty that the sizing procedure necessarily carries—most importantly the residual uncertainty in M , which is not known until the build is complete. A solid disc offers no such adjustment.

The design intent is that the bare printed ring supplies most of the required inertia and the ballast acts only as a trim, so that the bolt count sits well inside its feasible range in both directions.

Parameters to be fixed. The geometry is defined by the quantities of Table 18, all of which follow from the envelope closure of Section 5.2 and are recorded there once fixed.

Table 18: Reaction-wheel design parameters at the selected geometry. Ballast is standard M6 hex nuts (ISO 4032) in blind hexagonal pockets; m_b and ΔI_b are net of the plastic each pocket displaces.

Symbol	Parameter	Value
D_w	Ring outer diameter (capped by envelope)	130 mm
w_r	Ring radial width	20 mm
t_r	Ring axial thickness	12 mm
n_s, w_s	Spoke count and width	3, 10 mm
ρ_p	Printed-material density (PET-CF)	1290 kg/m ³
r_b	Ballast radius	55 mm
m_b	Net mass added per installed nut	1.87 g
ΔI_b	Net inertia added per installed nut	5.70×10^{-6} kg m ²
N	Nut count reaching $I_{w,\text{target}}$	15
m_w	Resulting wheel mass	156 g
I_w	Resulting wheel inertia	4.341×10^{-4} kg m ²
—	Achieved fraction of target	103.3 %

The bare ring and spokes supply 3.486×10^{-4} kg m², or 83 % of the target, so the ballast contributes the remaining 17 % — which is the intended proportion: enough that the trim is meaningful in both directions, not so much that the wheel depends on it.

The ballast is characterised *net* of the plastic each pocket displaces: m_b and ΔI_b are the increments the installed bolt-and-nut adds over the material it replaces, evaluated at the ballast radius. Sizing then reduces to solving

$$I_w(D_w, N) = I_{w,\text{ring}}(D_w) + N \Delta I_b \geq I_{w,\text{target}}$$

for the smallest admissible N at each candidate D_w , subject to the fit and mass checks below.

Trade study. Table 19 is the record sheet for the outer-diameter sweep, populated once $I_{w,\text{target}}$ and the envelope cap are known. Each candidate diameter is evaluated on four criteria, and the selected design is the one that passes all of them with the widest trim margin:

1. **Envelope.** D_w must clear the motor subframe, the motor body and the two orthogonal wheel modules inside the selected L .
2. **Inertia.** $I_w \geq I_{w,\text{target}}$, with margin for the uncertainty in M .
3. **Pocket fit.** Enough plastic wall must survive between adjacent pockets and radially inboard and outboard of each. Small diameters fail here: they need so many bolts to reach the target that no wall remains between them.
4. **Mass and trim margin.** The three-wheel set must stay within its allocation in Table 16. Diameters large enough that the bare ring already exceeds the target are rejected—they leave no downward trim authority and cost mass for nothing.

Criteria 3 and 4 bracket the feasible range from below and above respectively, so the selection is the largest diameter that still requires a non-trivial bolt count.

The trade brackets tightly. Below 130 mm every candidate fails criterion 3: reaching the target at a smaller radius needs so many nuts that the pockets overlap and no plastic wall survives between them — at 120 mm, 36 nuts on a 50 mm ballast circle leaves a negative wall. At 140 mm

Table 19: Reaction-wheel outer-diameter trade study against $I_{w,\text{target}} = 4.202 \times 10^{-4} \text{ kg m}^2$. N is the nut count reaching the target, rounded up to a multiple of the spoke count for balance; the wall column is the plastic left circumferentially between adjacent pockets against a 2 mm minimum.

D_w (mm)	N	Mass (g)	I_w (10^{-4})	I_w/target (%)	3 wheels (g)	Circ. wall
80	210	461	4.21	100.1	1382	Overlap
90	141	344	4.21	100.2	1031	Overlap
100	96	271	4.27	101.7	814	Overlap
110	60	216	4.22	100.5	648	Overlap
120	36	183	4.33	103.0	550	Overlap
130	15	156	4.34	103.3	468	11.0 mm OK
140	Bare ring exceeds target (4.51×10^{-4}) — no downward trim					
150	Bare ring exceeds target (5.73×10^{-4}) — no downward trim					

and above the bare ring already exceeds the target, failing criterion 4: the wheel costs mass for nothing and, more importantly, offers no downward trim authority if the measured inertia overshoots. 130 mm is the only diameter in the sweep that passes all four criteria, and it does so with 11.0 mm of circumferential wall — comfortably clear of the minimum rather than marginal.

Trim authority. The selected wheel sits at 103.3% of target with 15 nuts, and the pocket-placement limit below caps the ring at 21. The wheel can therefore be trimmed *upward* by up to six nuts ($+3.4 \times 10^{-5} \text{ kg m}^2$, about 8%) and downward by removing them. This is the margin the mass-budget risk of Section 5.2 draws on, and it is finite: it absorbs a moderate overrun in M , not an open-ended one.

Radial and floor fit. The radial checks pass in both nut orientations, which removes an assembly error mode — the nut fits either way round. A 12.01 mm pocket across corners in a 20 mm ring leaves 4.00 mm of wall each side at worst (4.80 mm across flats), against the 2 mm minimum. The pockets are blind rather than through-holes: a 5.2 mm pocket in the 12 mm ring leaves a 6.8 mm plastic floor beneath each nut, and that floor is a structural element rather than leftover material, since it is what reacts the centrifugal load.

Pocket placement. The pockets cannot be spaced uniformly, and this is a hard geometric result rather than a preference. A spoke occupies a finite angular width where it meets the ring, as does a pocket at the ballast radius, so a pocket centre must clear a spoke centre by the sum of the two half-widths. At the selected geometry the spoke half-angle at the ring inner diameter is 6.38° and the pocket half-angle at r_b is 6.27° , requiring 12.65° of separation. With $N = 15$ and three spokes the uniform pitch of 24° divides the spoke pitch exactly, placing every fifth pocket directly on a spoke root; offsetting by half a pitch does not rescue it either. No uniform pitch clears the spokes at any usable nut count.

The pockets are therefore distributed in equal groups within each inter-spoke sector, spread over the 94.7° of usable arc that remains once the spoke exclusion zones are removed. Grouping by sector preserves the three-fold symmetry and hence static and dynamic balance, and it caps the ring at seven pockets per sector — 21 in total, which is the source of the upward trim limit quoted above. This constraint is invisible to the sizing calculation, which models the ring as a bare annulus with no knowledge of where the spokes attach, so it is carried into the CAD as an explicit requirement (Appendix B.4) rather than left to be discovered when the wheel is drawn.

Ballast retention. Each nut carries a centrifugal load $F = m_b \omega_{\text{max}}^2 r_b = 54.3 \text{ N}$ (5.5 kg equivalent) at the design speed, reacted by the 6.8 mm pocket floor in bending. The floor will carry

it, but the floor alone is not a retention plan: each nut is a loose steel mass in a plastic pocket on a wheel whose rim moves at 40.8 m/s, and the load scales with $\omega_{\max}^2$, so any later decision to raise the design speed raises it quadratically.

Positive retention is therefore required and is provided by a retaining cover plate over the pocket mouths. This is chosen over adhesive or threadlocker specifically because it preserves the removability that motivates the entire ballast scheme — bonding the nuts in place would buy retention at the cost of the bidirectional trim that the mass-budget risk of Section 5.2 depends on. The retention method must be settled before the wheels are printed, since it affects the ring geometry.

Balance as an assembly-control problem. Fifteen discrete 2.5 g masses at 55 mm radius make wheel balance a matter of assembly verification, not just of design symmetry: a single missing or misplaced nut is a 2.5 g imbalance at that radius. Section 1.1 notes that a wheel half a gram out of balance at rim radius can render a correct estimator unusable, so nut-count and position verification per wheel is part of the balance and spin test rather than an inspection afterthought.

5.4 Electrical Design

The electrical architecture is presented here at the level of the four decisions that shape it: what the components are, how power is distributed, how the buses are routed and terminated, and how the whole is made safe to power on. The sheet-level schematics, the connector and pin tables, the harness drawings and the perfboard layout are collected in Appendix C, for the same reason the CAD detail is separated from the mechanical rationale — a connector re-pin should not require an edit here.

5.4.1 Components

The electronics and software will be built around a microcontroller communicating over a CAN bus with a motor driver that closes the current loop for the selected brushless motor. Wheel speed will be measured with a magnetic encoder on the motor shaft, and body tilt will be estimated from an onboard IMU. Braking will be actuated by a digital servo driven directly from the microcontroller, and a wireless module will provide a telemetry and debugging link. Power will be supplied by a LiPo battery, stepped down through a DC-DC converter for the logic-level electronics, with the motor driver powered directly from the battery bus.

The full parts list is catalogued in Appendix A (Table 25). The supplied set is an electronics-only BOM; the reaction wheels, the frame and all printed parts are the team’s design responsibility, and their fabrication is the leading schedule risk.

5.4.2 Power Architecture

The power architecture is shown in Figure 3. A single 6S LiPo pack (22.2 V nominal, 3600 mA h) feeds everything through one XT90 connector, behind a combined E-stop and fuse using an XT90-S anti-spark connector to limit inrush into the driver bulk capacitance. The bus then splits into two rails. The *motor bus* runs at pack voltage (22–25 V across the discharge curve) directly into the three moteus drivers, with bulk capacitance rated to at least 50 V to absorb regenerative braking transients—which are not incidental here, since the jump-up maneuver of Section 4.3 dumps the full wheel kinetic energy back into the bus. The *logic rail* is derived by an LM2596 buck converter stepping 22 V down to 5 V for the Teensy, the ESP32 telemetry link, the brake servo and the IMU. Keeping the two rails separate downstream of the E-stop means motor-current ripple does not reach the sensing and processing electronics.

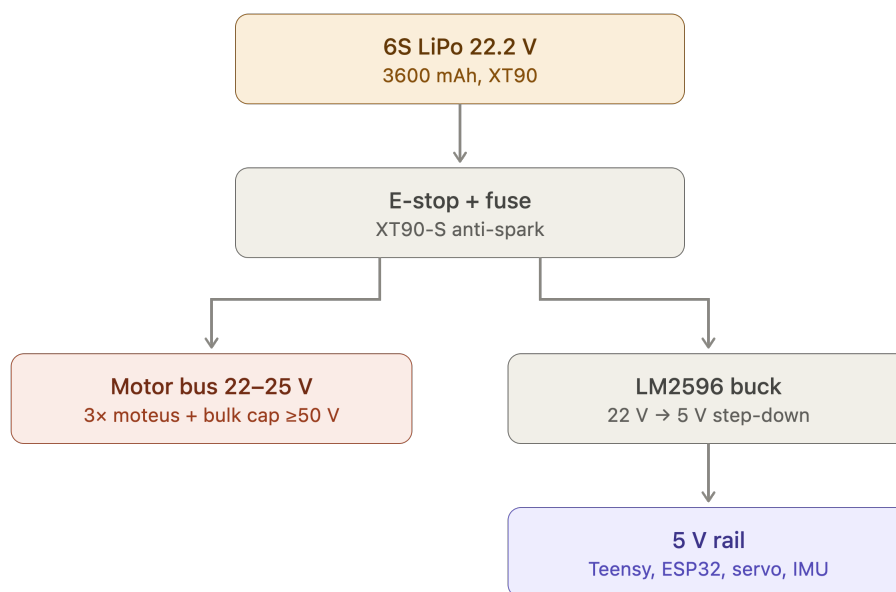


Figure 3: Power distribution tree. A single 6S pack feeds the motor bus directly and the 5 V logic rail through a buck converter, both behind a common E-stop and fuse.

The rail-by-rail current budget that sizes the buck converter and the harness conductors is Table 20. It is populated from measurement during bench bring-up rather than from datasheet maxima alone, because the figure that matters for the 5 V rail is the coincident worst case — telemetry transmit burst, servo stall and full sensor load at once — and not the sum of typical values.

Table 20: Rail current budget. Measured values replace estimates during bench bring-up; the 5 V total is checked against the LM2596 rating, which derates without a heatsink.

Load	Rail	Typical (mA)	Worst case (mA)	Basis
Teensy 4.1	5 V	TBD	TBD	Datasheet / measured
XIAO ESP32-C6 telemetry	5 V	TBD	300	TX burst governs
BMI270 IMU	5 V	TBD	TBD	Datasheet
MG92B brake servo	5 V	TBD	TBD	Stall current governs
MA600 encoders (3×)	5 V	TBD	TBD	Datasheet
Logic rail total	5 V	TBD	TBD	vs. 2–3 A converter rating
moteus-n1 drivers (3×)	Pack	TBD	TBD	Spin-up, not balance, governs

The logic-rail total is the figure to close first. The converter is rated 2–3 A and derates without a heatsink, while the coincident worst case — telemetry transmit burst at 300 mA, servo stall, and the full sensor load simultaneously — is not obviously inside that rating. Summing it from measurement is a bring-up task (Appendix C.6), not a design assumption.

The bulk capacitance on the pack bus is sized by the regenerative transient rather than by ripple: braking the wheels returns their stored energy (86 J per wheel at the design speed, Section 5.3.4) into the bus, so a 470–1000 μ F low-ESR part rated at 50 V or above is required. This is a distinct component from the small electrolytics on the logic rail, which are low-voltage parts and must

never be fitted to the pack bus — a confusion the isolation check of Appendix C.5 exists to catch.

5.4.3 Schematic and Interconnect Diagrams

The signal-level design is documented as a set of sheets rather than a single drawing, so that each can be checked against the constraint that governs it. Table 21 is the index; the sheets themselves are in Appendix C.1, together with the connector and pin-assignment tables that a person holding the harness needs and a reader of this section does not.

Table 21: Schematic sheet index. Each sheet is checked against the governing constraint stated in its row before release.

Sheet	Content	Governing constraint to verify
1	Power entry and protection	E-stop, fuse rating, anti-spark inrush into the driver bulk capacitance
2	5 V logic rail	Converter rating vs. Table 20 worst case; back-feed diode; rail bulk capacitance voltage rating
3	CAN-FD bus	Linear topology, stub length, split termination at the two physical ends only (Sec. 5.4.4)
4	Encoder interfaces (3×)	SPI cable length and encoder air gap; driver placement follows from this, not the reverse
5	Flight computer and IMU	Sensor orientation and mounting location relative to the pivot (Sec. 4.4)
6	Telemetry link	Transmit-burst current and its return path off the logic rail
7	Brake servo drive	Stall current headroom on the logic rail; PWM routing away from the encoder SPI

5.4.4 Bus Topology, Termination and Signal Integrity

The CAN-FD bus is the one electrical subsystem whose physical routing is a correctness requirement rather than a packaging preference, and it is therefore fixed before the board outline is drawn. At the 5 Mbit/s design rate the bus must be a linear chain with stubs under 10 cm and split termination ($2 \times 60.4 \Omega$ with a 4.7 nF capacitor to ground at the midpoint) at exactly the two physical ends; a star topology or a mid-chain termination will pass a bench continuity check and then fail intermittently under wheel vibration, which is the hardest class of fault to diagnose once the cube is closed. The design rules and their verification are listed in Table 22.

The encoder interfaces impose the complementary constraint, and it is the more restrictive of the two. Each MA600 requires an SPI run under 20 cm to its own driver and an air gap to its ring magnet held below 0.5 mm, with the sensor centred on the magnet thickness. The driver positions are therefore dictated by the encoder positions, and those in turn by the motor axes. The physical layout order is: motor axes fix the encoders, encoders fix the drivers, drivers fix the bus route, and only then is the board outline of Section 5.4.5 drawn around what remains. The shim value that sets the air gap is established on the bench and becomes an assembly specification carried into Appendix B.5, since the gap must be re-established every time the structure is bolted up.

5.4.5 Board Layout and Harness

The electronics are built on a 150 mm × 90 mm double-sided perfboard rather than a fabricated PCB. This is a schedule decision, not a technical preference: fabrication turnaround does not fit inside the build window, and a perfboard can be modified in place when bring-up reveals a change. Its consequence is that layout discipline — rail separation, decoupling at each device,

Table 22: Signal-integrity design rules and how each is verified. These are checked before cube closure, since all are hard to access afterwards.

Interface	Rule	Verification
CAN-FD bus	Linear chain; stubs < 10 cm; split termination ($2 \times 60.4 \Omega$, 4.7 nF) at the two physical ends only	Differential-pair scope capture; sustained error-frame count at 5 Mbit/s with all nodes addressed
Encoder SPI	Run < 20 cm to its own driver; air gap ≤ 0.5 mm, sensor centred on the magnet	Read-back with no error flags; re-checked after mechanical assembly
IMU	Mounted at or near the pivot; orientation per the body frame of Sec. 5.3.1	Static orientation check; noise floor with wheels spinning
Rail separation	Motor bus and logic rail separated downstream of the E-stop	Ripple measurement on the logic rail under motor load
Grounding	Single return topology; no motor return current through the sensing ground	Continuity and isolation check before first power-on

deliberate routing of the bus pair — must be enforced by hand and checked explicitly, since no design-rule check will do it.

The board dimension is also a driver of the cube envelope rather than a consequence of it: at 150 mm across, the board does not fit inside a 150 mm cube at any wall thickness, which is the packaging argument that independently forces the 180 mm envelope of Section 5.2.1. If the board can be split into two smaller boards that argument weakens considerably and the envelope decision rests more heavily on wheel trim authority; that question is settled during floorplanning and before the outline is delivered.

The board outline, mounting-hole pattern, keep-out heights and connector exit directions constitute the electronics-mount interface of Table 17 and are frozen once delivered to the mechanical design: later electrical changes must fit inside that outline rather than move it. The populated board mass is a required input to the mass budget of Table 16 and is obtained by weighing the assembled prototype, not estimated. The harness — battery lead, driver drops, bus chain and servo drive — is fabricated to the drawings in Appendix C.3 and is labelled and strain-relieved before installation, because every connector in the machine sits on a structure that vibrates by design.

5.4.6 Electrical Safety and Bring-Up Protocol

Two rules govern all electrical bring-up and are stated here because they are design constraints, not merely lab practice. First, every first-time power application is made on a current-limited bench supply, never on the battery; the battery is introduced only after the assembly has passed a full bench power-on. Second, the pack is never connected without the E-stop and fuse in circuit and the anti-spark path intact, since the driver bulk capacitance draws an inrush that will weld a bare connector.

The staged sequence — rails, then buses, then sensors, then open-loop motion, then closed-loop — is part of the validation plan and is given in Section 6. The isolation checks that precede first power-on (rail-to-rail separation, bus-to-chassis, and confirmation that no low-voltage capacitor sits on the pack bus) are recorded in Appendix C as a checklist, because they are performed on every re-assembly and not only once.

5.5 System Integration

5.5.1 Onboard Software Pipeline

The subsystems of Figure 2 are coordinated by a single deterministic task on the Teensy, whose structure is given in Figure 4. Each 1 ms cycle executes the same fixed sequence: sensor ingest reads the BMI270 over SPI and the moteus replies over CAN; the MEKF propagates attitude on the gyro and corrects it against the accelerometer while estimating gyro bias; the state vector $x = [g_B, \omega, h]$ is assembled from the estimator output and the wheel speeds; the LQR law of Section 4.4 evaluates $u = -Kx$ with a penalty on stored momentum h to bleed off wheel saturation; and the resulting commands leave as CAN torque requests to the drivers and a PWM signal to the brake servo. Telemetry is tapped off the assembled state to SD and Wi-Fi outside the control path, so logging cannot perturb loop timing.

The fixed ordering matters: with no dynamic allocation and no blocking calls, the worst-case cycle time is bounded by construction rather than measured after the fact, which is what makes the 1 kHz sample period of the discrete LQR design a design guarantee rather than an average.

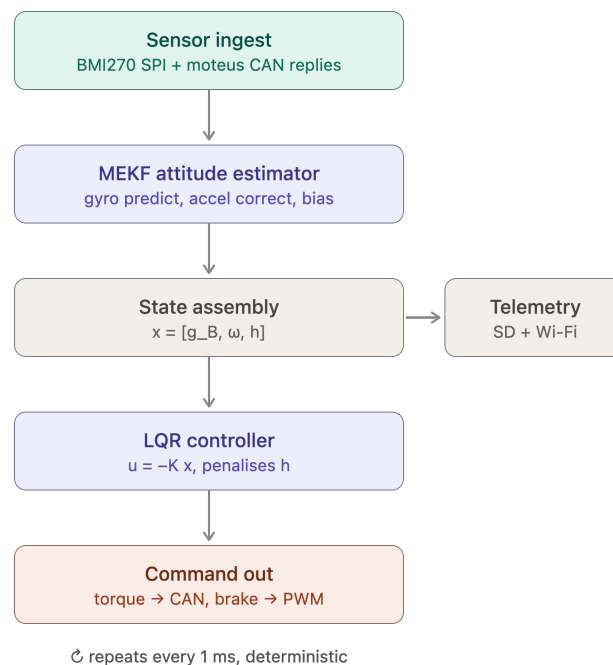


Figure 4: Teensy 4.1 control-loop software pipeline. The estimator and controller stages implement the design of Section 4.4; telemetry is a side-branch off the control path.

5.5.2 Control Software Architecture

Section 4 derives the control law; this subsection states how that law is realised as software, which is a separate set of decisions. The implementation detail — the mode state machine, the estimator and controller in executable pseudocode, the fixed gains and tuning parameters, the timing budget and the fault responses — is collected in Appendix D, so that a re-tuned gain does not require an edit to the design sections.

The software is organised in the four layers of Table 23. The division is not stylistic: each layer has a different failure mode and a different verification method, and the interfaces between them are what allow the controller validated in simulation to be the same code that runs on the cube.

Table 23: Control software layers. Each is verified by a different method, which is why the boundaries are held.

Layer	Responsibility	Verified by
Hardware abstraction	Sensor and driver I/O, timing, bus transactions	Bench read-back; loop-timing measurement
Estimation	Attitude and rate estimate from IMU and encoders (Sec. 4.4)	Static and dynamic bench tests against a known reference
Control	State feedback, momentum management, saturation handling	Simulation on the identified model, then hardware
Supervision	Mode transitions, arming, watchdog and fault response	Fault-injection tests on the rig

Operating modes. The machine is never in a single control regime. The supervisory layer sequences the modes of Table 24, with hysteresis on every threshold to prevent chattering at a boundary, and with a disarm transition available from every mode. Documenting the modes as a state machine rather than as a set of conditionals matters here because the dangerous transitions are the ones back down the ladder: a fall detected during corner balance must reach a safe state with the wheels commanded down, not merely switch to a different gain.

Table 24: Operating modes and transitions. Every mode has a disarm exit; the detailed conditions and hysteresis bands are in Appendix D.

Mode	Behaviour	Exit condition
Idle / disarmed	Drivers de-energised; sensors and telemetry live	Explicit arm command
Wheel spin-up	Open-loop speed ramp to the jump-up momentum target	Target wheel speed reached
Jump-up	Brake actuation and momentum transfer (Sec. 4.3)	Tilt inside the balancing capture band
Edge balance	Single-axis state feedback (Sec. 4.4)	Corner-transition command, or fall detected
Corner balance	Three-axis state feedback (Sec. 4.5)	Slew command, or fall detected
Slew / rotation	Reference tracking between equilibria	Reference reached; revert to balance
Fault / safe	Wheels commanded down, drivers disarmed	Manual reset

Real-time guarantees. The loop-rate claim of Section 4.4 is a design guarantee only if the worst-case cycle is bounded by construction, which is why the pipeline of Figure 4 uses a fixed stage ordering with no dynamic allocation and no blocking calls, and why telemetry is a side-branch rather than a stage. The per-stage timing budget and its measured margin are recorded in Appendix D and re-measured whenever a stage changes.

Fault handling. The failure cases the software must survive are sensor dropout, bus error, wheel saturation and loss of the balancing equilibrium. The first three have defined responses that keep the machine controlled; the fourth is a controlled shutdown rather than a recovery attempt, since a cube that has left its capture region cannot be recovered by the linear law and continued actuation only adds energy to the fall.

6 Testing and Validation Plan

This plan covers system-level validation. Parameter-identification tests that feed the model directly (swing test, current-step identification) are described with the model they validate in Section [4.2](#).

6.1 Simulation-First Validation

6.2 Bring-up Sequence

6.3 Sensor Testing

6.4 Milestone Acceptance Criteria

6.5 Control Performance Evaluation

7 Future Work

7.1 NMPC & Warm-Start Control (Future Implementation)

7.2 Autonomous Stand-up & Disturbance Rejection

7.3 Single-Wheel Variant / Extensions

References

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A Bill of Materials

The components supplied for this project constitute an *electronics* bill of materials: the actuation chain, wheel and attitude sensing, on-board compute, communications and power distribution are specified, but no structural or mechanical hardware is. The reaction wheels, the frame, the motor mounts, the fasteners and every printed bracket are the team’s design responsibility, and are treated as such in Section 5. Table 25 lists the supplied items grouped by subsystem. The actuation figures derived from these parts — the torque constant, the achievable torque and wheel speed, and the inertia that sizes the wheel — are developed where they are used, in Sections 4.3.1 and 5.3.4.

The architecture the BOM implies is a two-level control stack: each mjbots moteus-n1 closes a high-rate field-oriented current loop on its motor, using the MA600 encoder for rotor angle, while the Teensy 4.1 runs the outer attitude-estimation and state-feedback loop over CAN-FD — the same inner-tachometer / outer-attitude layering used in spacecraft reaction-wheel assemblies. A mechanical wheel brake, following the original ETH Cubli [2], provides the near-impulsive momentum transfer required for the jump-up.

Procurement is the single largest schedule risk. The long-lead items are not electronic but mechanical: the frame stock and the printed structure gate the entire build. Consumables and bench equipment not covered by the supplied set — wire, connectors, fasteners, PET-CF filament, the M6 ballast nuts, the bus bulk capacitor, XT30 pairs, the E-stop and fuse, the current-limited bench supply and the battery-safety kit — are tracked separately in the build documentation. The ballast nuts are ordered with generous surplus, since the wheel trim of Section 5.3.4 depends on having spares to hand.

No spare actuation set. Three complete motor, driver and encoder sets are procured for a three-axis machine, so there is no spare and resupply lead time does not fit inside the build window. Two consequences follow and neither is hypothetical. First, the 1-DoF bench rig of Section 5.3.2 draws on one of the three cube sets, so it cannot survive as a permanent development bench past cube integration — either it is stripped at that point or the cube runs on two axes. Second, the loss of any motor, driver or encoder in service permanently costs an axis, which would force the reduced-actuation one-wheel variant [8]. This is an accepted risk rather than a mitigated one.

Wheel procurement. The reaction wheels are printed PET-CF rings with M6 steel hex-nut ballast (Section 5.3.4). Any procurement line for cut steel rims belongs to a superseded wheel concept and is not part of this design.

Table 25: Supplied bill of materials, grouped by subsystem.

Ref	Item	Qty ^a	Key specifications	Function
Actuation				
1	T-Motor MN4006 KV380	3	18N24P outrunner; 380 rpm/V; 68 g; Ø44.35 mm × 21 mm; 194 mΩ; peak 16 A, 380 W	Reaction-wheel torque source
2	mjbots moteus- n1	3	10–54 V; 100 A peak / 9 A cont.; 15–30 kHz FOC; 5 Mbit/s CAN- FD; 14.6 g	FOC current/torque driver (inner loop)
11	TowerPro MG92B servo	1–3	metal-gear micro servo; ≈0.29 N m (≈3 kg cm) at 6 V; 50 Hz PWM	Wheel brake actuator (jump-up)

Continued on next page

Table 25 (continued)

Ref	Item	Qty ^a	Key specifications	Function
Sensing				
3	mjbots MA600 breakout + ring magnet	3	TMR absolute angle; SPI 6 MHz, 3.3 V; 16-bit (65,536 counts/rev); air-gap ≤ 0.5 mm	Wheel angle: commutation + state feedback
10	SparkFun BMI270 6-DoF IMU	1	3-axis accel + 3-axis gyro (no magnetometer); I ² C 400 kHz (Qwiic); ODR up to ≈ 1 kHz	Body attitude reference (tilt + rates)
Compute and Communications				
4	Teensy 4.1	1	600 MHz Cortex-M7 + FPU; native CAN-FD (CAN3); 5 V supply	Flight computer: estimator + control law
5	CAN-FD adapter (Teensy 4.1)	1	CAN-FD transceiver, logic-level $\rightarrow$ differential; 5 Mbit/s	Bus transceiver for the MCU
6	mjbots mjcanfd-usb-1x	1	USB $\leftrightarrow$ CAN-FD, 5 Mbit/s	Bench interface (calibration, tuning)
7	JST-PH3 CAN cables	≥ 4	TODO: conductor gauge / current rating not in source ; 3-conductor daisy-chain	CAN bus interconnect
8	Seeed XIAO ESP32-C6 + antenna	1	Wi-Fi; 3.3 V UART to MCU; TX burst ≈ 300 mA	Wi-Fi telemetry / command link
9	60.4 Ω 0.5 W resistor	4	split termination, $2 \times 60.4 \Omega \approx 120 \Omega$ bus impedance	CAN bus termination
17	4.7 nF capacitor	2	common-mode filter, bus midpoint to ground	CAN split-termination filter
Power				
12	Turnigy 6S LiPo (XT90)	1	22.2 V nom / 25.2 V charged; 3600 mA h; 60C; ≈ 80 W h; ≈ 600 g	Primary energy store
13	LM2596 buck module	1	step-down to 5.0 V; rated 2–3 A (derates without heatsink)	5 V logic rail
14	XT90 connector pair	1–2	rated ≈ 90 A	Battery power connector
15	100 μ F 16 V electrolytic	2–4	16 V part — 5 V rail only	5 V-rail bulk reservoir
19	1N5819 Schottky diode	1–2	1 A, 40 V	5 V-rail back-feed protection
Integration				
16	100 nF 50 V capacitor	6–10	local decoupling at each IC	Per-IC supply decoupling
18	Double-sided perfboard	1	15 cm $\times$ 9 cm, double-sided	Power / signal distribution board

^aQuantity per assembled cube; motors, drivers and encoders are 3 each, with *no spare*. The 1-DoF bench rig therefore uses one of the three cube actuation sets and must be stripped at cube integration. See the procurement note below.

B CAD and Construction Detail

This appendix holds the mechanical detail deliberately kept out of Section 5.3: the model tree, the drawing set, the fabrication settings and the assembly sequence. The rationale for the architecture — why the frame is split, what fixes the envelope, how the wheel is tuned — stays in Section 5.3 and is not repeated here. Anything in this appendix may change without the design intent changing; anything in Section 5.3 may not.

B.1 Design Parameter Table

Table 26 is the single source of the driving dimensions referenced by every part in the model, per convention 1 of Section 5.3.1. It is the CAD-side mirror of the envelope closure of Section 5.2 and is updated on each iteration.

Table 26: Driving CAD parameters. Every dependent feature references these; no part carries a duplicated hard-coded value.

Symbol	Parameter	Value
L	Cube edge length (outer)	180 mm
D_w	Wheel outer diameter (Sec. 5.3.4)	130 mm
w_r	Wheel ring radial width	20 mm
t_r	Wheel ring axial thickness	12 mm
r_b	Ballast pocket circle radius	55 mm
—	Board outline (Sec. 5.4.5)	150 mm × 90 mm
t_w	Frame wall thickness	TBD
d_m	Motor axis offset from cube centre	TBD
h_m	Motor mounting face height on its axis	TBD
c_w	Wheel-to-subframe axial clearance	TBD
c_o	Clearance between orthogonal wheel modules	TBD
$V_{\min}$	Minimum internal clear volume	TBD
r_c	Corner contact feature radius	TBD

Open input. The frame wall thickness and internal clearance that set the “ L minus 30–40 mm” wheel-diameter allowance used in Table 15 are an estimate rather than a CAD result. The envelope argument holds anywhere in that band, but the selected ring diameter depends on where in it the real frame lands, so the maximum wheel diameter that actually fits the 180 mm frame must be confirmed from the model before the ring is frozen.

B.2 Model Tree and Part List

Table 27 lists the modelled parts by assembly, with their fabrication route and current status. It is the mechanical counterpart to the electronics bill of materials of Appendix A: the parts here are the team’s design responsibility and are not supplied.

B.3 2D Prototype Construction

B.4 3D Cube Construction

Wheel pocket placement — a CAD constraint the sizing cannot check. The ballast pockets must be drawn in three groups of five within the inter-spoke sectors, *not* at a uniform angular pitch. The reasoning and the angular figures are in Section 5.3.4; what matters here is that the sizing calculation models the ring as a bare annulus with no knowledge of the spokes,

Table 27: CAD model tree and part list. Quantities are per assembled cube unless noted; the 1-DoF rig parts are listed separately.

Ref	Part	Qty	Fabrication route	Status
Inner structure				
M1	Motor subframe	TBD	TBD	TBD
M2	Motor mount ($\times$ axis)	3	TBD	TBD
M3	Battery cradle / retention	1	TBD	TBD
M4	Electronics tray	1	TBD	TBD
M5	Brake servo bracket	3	TBD	TBD
Reaction wheel assembly				
W1	Ballasted ring, 130 mm OD	3	Printed PET-CF	Geometry fixed
W2	Wheel hub / rotor adapter	3	Printed	TBD
W3	M6 hex nut ballast (ISO 4032)	45	Standard hardware	15 per wheel
W4	Ballast retaining cover	3	TBD	Required, not yet designed
W5	Brake strike feature	3	TBD	TBD
Outer frame				
F1	Frame member / edge rail	TBD	TBD	TBD
F2	Corner contact feature	8	TBD	TBD
F3	Face panel / wheel guard	TBD	TBD	TBD
1-DoF rig (Sec. 5.3.2)				
R1	Face plate	1	TBD	TBD
R2	Pivot bearing mount	1	TBD	TBD
R3	Rig motor mount	1	TBD	TBD
R4	Hard stops / containment	TBD	TBD	TBD

so it will report a feasible design for a uniform layout that in fact places pockets on the spoke roots. The layout rule is Table 28 and it governs the drawing.

B.5 Fabrication and Assembly

B.5.1 Print and Manufacturing Settings

The printed parts are structural, not cosmetic: the wheel ring carries centrifugal load at the ballast pockets and the motor mounts carry the brake impulse, which is the design load case identified in Section 1.1. Settings are therefore recorded per part in Table 29 and treated as part of the design, since a part reprinted at a different orientation or infill is not the part that was analysed.

The wheel ring density is not a free choice at print time: the inertia figures of Section 5.3.4 assume 1290 kg/m^3 throughout, so a ring printed at reduced infill supplies less inertia than designed and moves the nut count. Print settings for W1 are therefore part of the sizing, and a re-print at different settings requires the ballast count to be re-derived.

Table 28: Ballast pocket layout rule. Grouping by sector preserves three-fold symmetry and caps the ring at 21 pockets.

Quantity	Value
Spoke half-angle at ring inner diameter	6.38°
Pocket half-angle at $r_b = 55$ mm	6.27°
Required spoke-centre to pocket-centre separation	12.65°
Usable arc per sector, after spoke keep-outs	94.7°
Pockets per sector at $N = 15$	5
Maximum pockets per sector	7 (21 total)

Table 29: Fabrication settings for the structural printed parts. Layer orientation is specified relative to the principal load direction.

Ref	Material	Infill / walls	Orientation and rationale
W1	PET-CF, 1290 kg/m ³	TBD	Layers normal to the spin axis, so the hoop load and the ballast pocket walls are not carried across a layer boundary
W2	TBD	TBD	Carries the brake impulse into the rotor; orientation set by that load path
M1	TBD	TBD	TBD
M2	TBD	TBD	Carries the brake reaction; the governing load case of Sec. 1.1
F1	TBD	TBD	TBD
F2	TBD	TBD	TBD

B.5.2 Assembly Sequence

The sequence of Table 30 is ordered by access: each step is one whose verification becomes impossible or destructive once a later step is complete. The encoder air gap in particular is re-checked after mechanical assembly, since it can shift when the structure is bolted up (Table 22).

B.6 Mass Properties Verification

The CAD roll-up is an estimate until the assembly is weighed. Step 8 above closes the loop of Section 5.2: the measured mass is compared against the modelled value, and any discrepancy is carried back into $I_{w,\text{target}}$ and absorbed, where possible, by the ballast trim of Section 5.3.4 rather than by a re-design.

Two of these rows are gating rather than informational. The measured M decides whether the mass budget of Section 5.2 closes at all, and it propagates through $h_w \propto M$ into the wheel; the measured nut mass propagates into I_w directly. On the latter, the design uses a measured 2.5 g against a geometric estimate of 2.617 g for an M6 nut to ISO 4032 — a -4.5% discrepancy consistent with ordinary manufacturing tolerance and plating variation, but worth re-confirming on a scale before the nut count is frozen, since the shape (and hence the radius of gyration) is taken from the nominal geometry even when the mass is measured.

Table 30: Assembly sequence. Ordering is set by access: each check must be completed before the step that encloses it.

Step	Operation	Check before proceeding
1	Wheel assembly: ring, hub, ballast bolts to the selected count	Balance check; bolt retention per Sec. 5.3.4
2	Motor to mount, encoder magnet to rotor	Encoder air gap and centring to the assembly spec
3	Motor modules to inner structure, three axes	Axis orthogonality; wheel clearance c_w , c_o
4	Battery and electronics tray into inner structure	Pack centring; board keep-out clearances
5	Harness install and dressing	Continuity and isolation (Sec. 5.4.6)
6	Inner structure into outer frame	Encoder air gaps re-checked after bolt-up
7	Contact features and guards	Contact geometry; wheel guard clearance
8	Mass properties measurement	Measured mass against Table 16

Table 31: Modelled versus measured mass properties. A discrepancy here is absorbed by ballast trim where the margin allows.

Quantity	Design value	Measured	Discrepancy
Total cube mass M	1400 g	TBD	TBD
Wheel mass m_w (each)	156 g	TBD	TBD
Wheel inertia I_w (each)	4.341×10^{-4}	TBD	TBD
Ballast nut mass (each)	2.5 g	TBD	TBD
CoM offset from cube centre	0 (by design)	TBD	TBD

C Electrical System Detail

This appendix holds the electrical detail kept out of Section 5.4: the schematic sheets indexed in Table 21, the connector and pin assignments, the harness drawings, the board layout and the bring-up checklists. The architectural decisions — two rails behind one E-stop, linear bus topology, perfboard rather than fabricated board — remain in Section 5.4. The part specifications are in Appendix A and are not repeated.

C.1 Schematic Sheets

C.1.1 Sheet 1 — Power Entry and Protection

C.1.2 Sheet 2 — Logic Rail

C.1.3 Sheet 3 — CAN-FD Bus

C.1.4 Sheet 4 — Encoder Interfaces

C.1.5 Sheet 5 — Flight Computer and IMU

C.1.6 Sheet 6 — Telemetry Link

C.1.7 Sheet 7 — Brake Servo Drive

C.2 Connector and Pin Assignments

Table 32 is the connector schedule. It exists so that a harness can be built and re-built from this document alone, and so that a mis-mate is a documented impossibility rather than a discovered one: no two connectors carrying different voltages share a housing type on this machine.

Table 32: Connector schedule. Keying and housing types are chosen so that two connectors carrying different voltages cannot be mated.

Ref	From	To	Housing / keying	Signals
J1	Battery pack	Protection / E-stop	XT90	Pack +, pack –
J2	Protection	Distribution board	XT90	Motor bus
J3	Distribution board	Driver, axis 1	XT30	Motor bus
J4	Distribution board	Driver, axis 2	XT30	Motor bus
J5	Distribution board	Driver, axis 3	XT30	Motor bus
J6	Board	CAN chain, end A	JST-PH3	CAN-H, CAN-L, GND
J7	CAN chain, end B	Termination network	JST-PH3	CAN-H, CAN-L, GND
J8	Driver, axis i	Encoder, axis i	TBD	SPI, 3.3 V, GND
J9	Board	IMU	Qwiic	I ² C, supply, GND
J10	Board	Telemetry module	TBD	UART, supply, GND
J11	Board	Brake servo, axis i	2.54 mm header	PWM, 5 V, GND

Note the connector mismatch at J2–J5: the pack terminates in XT90 while the drivers accept XT30, so a distribution harness must be fabricated rather than bought. It is the item H3 of Table 33 and it is on the critical path for first power-on.

C.3 Harness Drawings and Wire Schedule

Table 33 is the wire schedule. Conductor gauge on the motor bus is set by the spin-up current rather than the balancing current, since spin-up is the sustained high-current phase; the signal runs are set by length limits rather than by current.

Table 33: Wire schedule. Motor-bus gauge is sized on spin-up current; signal runs are length-limited (Table 22).

Ref	Run	Gauge	Length	Sizing basis
H1	Pack to protection	14 AWG	TBD	Peak bus current
H2	Protection to board	14 AWG	TBD	Peak bus current
H3	Board to driver drops (3)	14 AWG	TBD	Per-axis spin-up current
H4	Motor phase leads (3)	TBD	TBD	Motor phase current
H5	CAN chain segments	TBD	stubs < 10 cm	Topology, not current
H6	Encoder SPI runs (3)	24–26 AWG	< 20 cm	Length limit governs
H7	Servo drives (3)	24–26 AWG	TBD	Stall current

The conductor gauge of the supplied three-conductor CAN cable is not stated by its supplier, so H5 is an open item: whether stock cables can be used or the bus harness must be hand-made is decided by that figure, and it affects the floorplan only marginally but the build time appreciably.

C.4 Board Layout

The floorplan is constrained before it is aesthetic: the driver positions follow from the encoder cable limit, the bus route follows from the driver positions, and the board is placed around what remains (Section 5.4.4). Rail separation is maintained physically on the board, not only schematically, so that motor-current return does not share copper with the sensing ground.

C.5 Bring-Up and Isolation Checklists

These checks are performed on every re-assembly, not only on first build, since the encoder gaps and connector seating are exactly what mechanical work disturbs.

Before any power is applied.

1. Isolation: motor bus to chassis, logic rail to motor bus, each rail to ground.
2. Capacitor voltage ratings confirmed by position — no low-voltage part on the pack bus.
3. Polarity of every fabricated connector verified against Table 32.
4. Protection element in circuit: E-stop functional, fuse fitted and rated, anti-spark path intact.
5. Encoder air gaps re-measured after mechanical assembly.

First power-on, current-limited bench supply.

1. Current limit verified against a known load before connecting the assembly.
2. Logic rail voltage and ripple under load; converter thermal check.
3. All drivers enumerate on the bus; encoders read back with no error flags; IMU responds.

4. Bus integrity at the full data rate: sustained soak with all nodes addressed, error-frame count logged.
5. Open-loop wheel motion, wheels contained, one axis at a time.

First battery power-on. Performed only after the bench sequence passes in full, with the pack protection verified, cell monitoring active and the assembly in a fire-safe area. Inrush is measured against the bulk capacitance on connection.

C.6 As-Built Measurements

Table 34 records the measured electrical figures that replace the estimates used during design. These are the numbers reported back to the modelling and control work, and they are the reason the parameter identification of Section 4.2 runs on measured rather than assumed values.

Table 34: As-built electrical measurements. Each replaces an estimate used earlier in the design.

Quantity	Design value	Measured	Replaces / feeds
Motor torque constant K_m	0.025,13 N m/A	TBD	Table 6, datasheet value
Phase resistance	194 m Ω	TBD	Incoming inspection; speed ceiling
Wheel inertia I_w (spin-down)	4.341×10^{-4}	TBD	Table 6; ballast count
Viscous and Coulomb friction	Estimated	TBD	C_b , C_w estimates
Command-to-encoder latency	—	TBD	Loop timing, Sec. 5.5.2
Achievable wheel speed on bus	6000 rpm	TBD	$\omega_{\max}$, Sec. 4.3.1
Brake torque τ_b	5.0 N m (assumed)	TBD	β , Sec. 4.3.2
Logic rail load, worst case	TBD	TBD	Table 20
Populated board mass	TBD	TBD	Table 16
Runtime per charge, balancing	—	TBD	Demonstration planning

Two rows here gate design decisions rather than merely confirming them. The brake torque is the unsubstantiated input identified in Section 4.3.2, and it must be measured on the 1-DoF rig before the wheel geometry is frozen, since β propagates straight into the inertia target. The achievable wheel speed is checked against the design value on a clamped motor during single-axis bring-up: if the measured ceiling falls below 6000 rpm on a partly-discharged pack, the conservatism argument of Section 4.3.1 has not been realised and the wheel is undersized rather than over-margined.

D Control Algorithms

This appendix holds the implementation detail of the control software whose architecture is given in Section 5.5.2. The derivations remain in Section 4 and are not repeated; what follows is the executable form — the loop, the estimator, the control law, the mode logic and the fault responses — together with the gains and timing figures that are re-tuned against hardware and therefore change more often than the design does.

D.1 Control Loop

The top-level loop is the fixed sequence of Figure 4, executed at the sample period Δt of Table 11. Its ordering is a real-time requirement, not a convenience: with no dynamic allocation and no blocking call, the worst-case cycle is bounded by construction.

```
every dt:
    t0 <- now()

    # 1. Sensor ingest -- fixed-size, non-blocking reads only
    imu      <- read_imu()           # rates and specific force
    wheels   <- read_driver_replies() # wheel angle and speed, per axis
    flag_stale(imu, wheels)          # feeds the fault logic below

    # 2. Estimation -- propagate on rates, correct on the absolute reference
    xhat     <- estimator_predict(xhat, imu.rates, dt)
    xhat     <- estimator_correct(xhat, imu.accel)

    # 3. State assembly
    x        <- assemble_state(xhat, wheels)

    # 4. Supervision -- mode first, so the control law below is unambiguous
    mode     <- supervisor_update(mode, x, faults)

    # 5. Control law for the current mode
    u        <- control_law(mode, x, gains[mode])
    u        <- apply_limits(u)       # torque and momentum saturation

    # 6. Output
    send_torque_commands(u)
    set_brake(mode)

    # 7. Telemetry -- side-branch, never in the control path
    telemetry_publish_nonblocking(x, u, mode)

    log_cycle_time(now() - t0)       # feeds the timing budget
```

Telemetry sits after the output stage and is non-blocking by construction, so that logging cannot perturb loop timing — the property that makes the sample period a guarantee rather than an average.

D.2 Attitude Estimation

The estimator propagates attitude on the measured rates and corrects it against the absolute reference, while estimating the rate-sensor bias. The two error sources it must handle pull in

opposite directions and are the reason a single sensor will not do: the rate measurement is clean over a cycle but its bias drifts, and the absolute reference is drift-free but is corrupted by the vibration the machine itself generates by spinning the wheels (Section 1.1).

```
estimator_predict(xhat, rates, dt):
    w_corrected <- rates - xhat.bias
    xhat.att     <- integrate_attitude(xhat.att, w_corrected, dt)
    xhat.P       <- propagate_covariance(xhat.P, dt)    # bias random walk
    return xhat

estimator_correct(xhat, accel):
    # Gate the correction: during high-acceleration phases the absolute
    # reference is not measuring the vertical, and applying it would inject
    # the manoeuvre into the attitude estimate.
    if not accel_trustworthy(accel):
        return xhat
    innovation <- reference_direction(accel) - predicted_direction(xhat.att)
    K          <- gain(xhat.P, R_accel)
    xhat       <- apply_correction(xhat, K, innovation)    # attitude and bias
    return xhat
```

The gating in `estimator_correct` is the load-bearing part. The jump-up and any aggressive recovery are exactly the phases in which the accelerometer does not measure the vertical, so an ungated correction would drive the estimate with the manoeuvre. The trust test and its threshold are recorded in Table 37.

D.3 Control Law

The balancing law is the state feedback of Sections 4.4 and 4.5, with two additions required by hardware: the stored-momentum penalty that bleeds off wheel speed, and the offset filter that absorbs a constant tilt-sensor bias without a calibration stage.

```
control_law(mode, x, K):
    if mode in {EDGE_BALANCE, CORNER_BALANCE, SLEW}:
        e      <- x.att - reference(mode)    # zero for pure regulation
        x_f    <- lowpass(x_f, e)            # absorbs constant sensor offset
        u      <- -K @ [e - x_f, x.rates, x.wheel_speeds]
        return u

    if mode == SPINUP:
        return speed_ramp(x.wheel_speeds, target_speed)

    if mode == JUMPUP:
        return open_loop_jump_sequence(t_since_mode_entry)

    return zero_torque()                    # IDLE and FAULT
```

The wheel-speed term in the feedback is not merely a damping state. Under any persistent disturbance the wheels drift toward their speed limit, so the gain on stored momentum is what keeps the actuators inside their envelope; without it the machine balances correctly until it saturates and then does not.

Saturation handling. `apply_limits` enforces the torque limit and the momentum envelope. Torque is clipped per axis; momentum saturation is handled by the speed penalty above rather

than by clipping, since clipping a saturated wheel removes control authority precisely when it is needed. When a wheel reaches its limit despite the penalty, the supervisor is notified and treats it as the fault of Section D.5.

D.4 Mode Supervision

The supervisor implements the transitions of Table 24. Every threshold carries hysteresis, and every mode has a disarm exit.

```
supervisor_update(mode, x, faults):
    if faults.critical:
        return FAULT                # from any mode, unconditionally

    switch mode:
        IDLE:                if arm_command:                -> SPINUP
        SPINUP:               if wheel_speed >= target:     -> JUMPUP
        JUMPUP:               if |tilt| < capture_band:     -> EDGE_BALANCE
                           if timeout:                     -> FAULT
        EDGE_BALANCE:         if |tilt| > release_band:     -> FAULT        # fall detected
                           if corner_command:              -> CORNER_BALANCE
        CORNER_BALANCE:       if |tilt| > release_band:     -> FAULT
                           if slew_command:                -> SLEW
        SLEW:                 if reference_reached:         -> CORNER_BALANCE
                           if |tilt| > release_band:       -> FAULT
        FAULT:                if manual_reset and safe:    -> IDLE
    return mode
```

The capture and release bands differ deliberately — entry into balancing is stricter than the condition for staying in it — which is the hysteresis that prevents chattering at a mode boundary.

D.5 Fault Detection and Response

Table 35 gives the detection test and response for each failure case. The distinction that matters is between faults the machine can continue through and faults it cannot: only the loss of the balancing equilibrium terminates the run, because a body outside its capture region cannot be recovered by the linear law, and continued actuation adds energy to a fall that is already happening.

D.6 Timing Budget

Table 36 records the per-stage timing budget and the measured margin. It is re-measured whenever a stage changes, because the guarantee of Section 5.5.2 is a claim about the worst case and not about the mean.

D.7 Gains and Tuning Parameters

Table 37 is the tuning record: every constant the control software carries, with the source that fixes it. Values computed from the model are distinguished from values tuned against hardware, since only the latter must be re-established when the machine is re-built.

Table 35: Fault detection and response. Only loss of equilibrium terminates the run; the remaining cases keep the machine controlled.

Fault	Detection	Response
Sensor dropout / stale data	Missing or repeated sample within the cycle deadline	Propagate on the remaining sensor; escalate to FAULT if persistent beyond the tolerance
Bus error	Error-frame count or missed driver reply	Retry within the cycle; escalate on repetition
Encoder error flag	Flag set, or angle discontinuity beyond the physical rate limit	Disarm the affected axis; FAULT if it is load-bearing for the current mode
Wheel saturation	Speed at limit despite the momentum penalty	Notify supervisor; reduce reference authority; FAULT if sustained
Loss of equilibrium	Tilt outside the release band	FAULT: wheels commanded down, drivers disarmed
Loop overrun	Measured cycle time exceeds the deadline	Log; FAULT if repeated, since the timing guarantee no longer holds
Watchdog	Loop fails to service the timer	Hardware disarm, independent of the software path

Table 36: Per-cycle timing budget against the sample period. Re-measured whenever a stage changes.

Stage	Budget	Measured worst case
Sensor ingest	TBD	TBD
Estimation	TBD	TBD
State assembly	TBD	TBD
Supervision	TBD	TBD
Control law	TBD	TBD
Command output	TBD	TBD
Total cycle	TBD	TBD
Margin to deadline	—	TBD

Table 37: Control and estimation parameters. “Model” values follow from the design; “bench” values are tuned against hardware and re-established after a re-build.

Symbol	Parameter	Source	Value
Δt	Sample period	Model	TBD
Q, R	State and input weights	Model	TBD
K_d (edge)	Feedback gain, edge balance	Model	TBD
K_d (corner)	Feedback gain, corner balance	Model	TBD
—	Momentum-penalty weight	Bench	TBD
γ	Estimator correction weight	Bench	TBD
—	Accelerometer trust threshold	Bench	TBD
α	Offset-filter coefficient	Bench	TBD
θ_{th}	Balancing capture band	Bench	TBD
—	Balancing release band (hysteresis)	Bench	TBD
—	Spin-up target wheel speed	Model	TBD
—	Brake actuation timing	Bench	TBD
—	Torque limit per axis	Model	TBD
—	Watchdog timeout	Model	TBD